

Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2024.0429000

From Hand-Drawn Schematics to SPICE Netlists: A Deterministic Pipeline with Endpoint-Graph Wire Joining and a Human-Verified Connectivity Benchmark

BOSCO CHANAM¹ , CHRIS DCOSTA¹ , PRANAVESH KUMAR TALUPURI² , SHWETAMBARI CHIWHANE¹ , ASHAY KUMAR SINGH¹ , and ARGHADEEP DAS¹ 

¹Symbiosis Institute of Technology, Pune 412115, India (e-mail: boscochanam@gmail.com; chrisdcosta777@gmail.com; shwetambari.chiwhane@sitpune.edu.in; ashay.sit@gmail.com; arghadeep.das.work@gmail.com)

²Department of Computer Science, University of Southern California, Los Angeles, CA 90089 USA (e-mail: talupuri@usc.edu)

Corresponding author: Bosco Chanam (e-mail: boscochanam@gmail.com).

This work received no external funding.

ABSTRACT Automated digitization of hand-drawn circuit schematics into simulation-ready SPICE netlists remains challenging because connectivity recovery is brittle under noisy wire detections. We present a deterministic pipeline that converts a scanned schematic into a SPICE netlist through component detection, an occlusion-first wire extractor, and a graph-based join that resolves electrical connectivity. The main contribution is the joining stage: an endpoint-graph model over wire endpoints and component pins with five edge types, combined with degree-budget completion that recovers dropped connections through min-cost b-matching while bounding over-merge with per-pin budgets and self-loop guards. We also introduce a synthetic perturbation protocol and, to our knowledge, the first human-verified net-level connectivity benchmark for hand-drawn circuits, covering 31 images from CGHD-1152. The occlusion-first extractor reaches wire-detection $F1 = 0.976$ on 134 scanned images. On the human-verified net-level benchmark, the best join reaches connectivity micro- $F1 = 0.890$, outperforming the prior completion default (0.83), a legacy radius union-find join (0.67), a connected-component net-tracing baseline (0.62), and a Hough-plus-proximity line tracer (0.81), all under identical detected wires. Running the join on perfect wire labels does not improve micro- $F1$, showing that connectivity, not wire detection, remains the bottleneck. A first-party comparison shows a strong vision-language model reaches 0.923 on the same images, statistically indistinguishable from our join but far more expensive and without structural guarantees, supporting explicit geometric connectivity recovery for deployable circuit digitization.

INDEX TERMS circuit digitization, connectivity benchmark, graph-based connectivity, hand-drawn circuits, image-to-netlist, netlist extraction, schematic recognition, SPICE, wire extraction, wire joining

I. INTRODUCTION

Analog circuit schematics encode functional information (component types, values, and topological connections) in a visual format designed for human interpretation. Converting scanned schematics into machine-readable SPICE (Simulation Program with Integrated Circuit Emphasis) netlists enables automated simulation, design verification, and large-scale circuit analysis. While re-

cent advances in object detection have made component recognition increasingly reliable, the joining problem, determining which wire segments connect to which component pins and to each other, remains the primary failure mode in end-to-end digitization pipelines.

The challenge is threefold. First, wire endpoints extracted from binarized images are inherently noisy: the same circuit drawn by different drafters produces

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Attribution note: Pranavesh authored earlier benchmark commits. Git lists Bosco Chanam as author of the later manuscript revision commits; confirm individual edit credit with the authors.

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From Hand-Drawn Schematics to Structural Circuit Netlists: A Deterministic Pipeline with Endpoint-Graph Wire Joining and a Human-Verified Connectivity Benchmark

BOSCO CHANAM¹ , CHRIS DCOSTA¹ , PRANAVESH KUMAR TALUPURI² , SHWETAMBARI CHIWHANE¹ , ASHAY KUMAR SINGH¹ , and ARGHADEEP DAS¹ 

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²Department of Computer Science, University of Southern California, Los Angeles, CA 90089 USA (e-mail: talupuri@usc.edu)

Corresponding author: Bosco Chanam (e-mail: boscochanam@gmail.com).

This work received no external funding.

ABSTRACT We present a deterministic pipeline for structural circuit-netlist recovery using an occlusion-first wire extractor, an endpoint graph, and degree-budget completion. The output represents inferred pin-to-node assignments; our component-pair metric does not certify pin identities, exact net partitions, absence of shorts, or simulation equivalence. Illustrative SPICE export requires externally specified values and device models; simulation equivalence on real scans is not validated. We provide a synthetic perturbation protocol and a human-verified connectivity benchmark of 31 CGHD-1152 images. With annotated component boxes and detected wires, the join achieves component-pair micro-F1 0.890 (macro-F1 0.901), compared with 0.805 for Hough plus proximity and 0.624 for connected-component tracing on the same detected wires. The wire extractor achieves F1 0.976 on 134 images with annotated component boxes under the historical benchmark configuration. Under the same component priors on the 31-image benchmark, a vision-language model reaches micro-F1 0.923; the paired difference is +0.033, with 95% confidence interval $[-0.009, +0.078]$. Inclusion of zero does not establish equivalence or comparative cost superiority. These results provide within-corpus evidence, not validation of autonomous reconstruction or unseen-style generalization.

INDEX TERMS circuit digitization, connectivity benchmark, graph-based connectivity, hand-drawn circuits, image-to-netlist, netlist extraction, schematic recognition, SPICE, wire extraction, wire joining

I. INTRODUCTION

Analog circuit schematics encode component types, values, and topological connections in a visual format designed for human interpretation. Recovering a structural netlist from a scan is a prerequisite for simulation, design verification, and large-scale circuit analysis, but a complete simulatable SPICE (Simulation Program with Integrated Circuit Emphasis) deck also requires component values and device models. This work addresses the structural connectivity stage only. The joining problem determines which wire segments connect to which component pins and to each other. We study this stage with annotated component boxes; its relative contribution to autonomous reconstruction error is not established by that experiment.

The challenge is threefold. First, wire endpoints extracted from binarized images are inherently noisy: the same circuit drawn by different drafters produces endpoint positions that vary by 10–80 pixels. Second, components occupy bounding boxes that overlap with wire endpoints, creating ambiguity about whether an endpoint connects to a component’s pin or merely passes nearby. Third, T-junctions, collinear fragments, and crossing wires create connectivity ambiguities for proximity-based rules, including the graph rules studied here.

A complementary approach is to prompt a vision-language model (VLM) for connectivity. Section IV-H reports an oracle-component-box diagnostic in which both methods receive the same annotated component boxes and the geometric method uses detected wires. The observed component-pair

GOLD = added / revised text

Title, abstract, and claim boundary

Structural-netlist title replaces SPICE-ready wording.

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Git author on manuscript revisions: Bosco Chanam; individual contribution credit requires confirmation.

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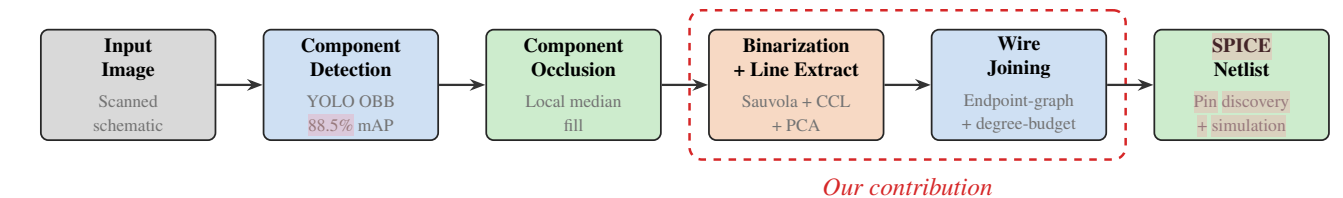


FIGURE 1. End-to-end pipeline for scanning circuit diagram digitization. Our contributions span the wire extraction and joining stages; the join (dashed red) determines which extracted wire segments connect to which component pins and is the dominant failure mode in end-to-end digitization.

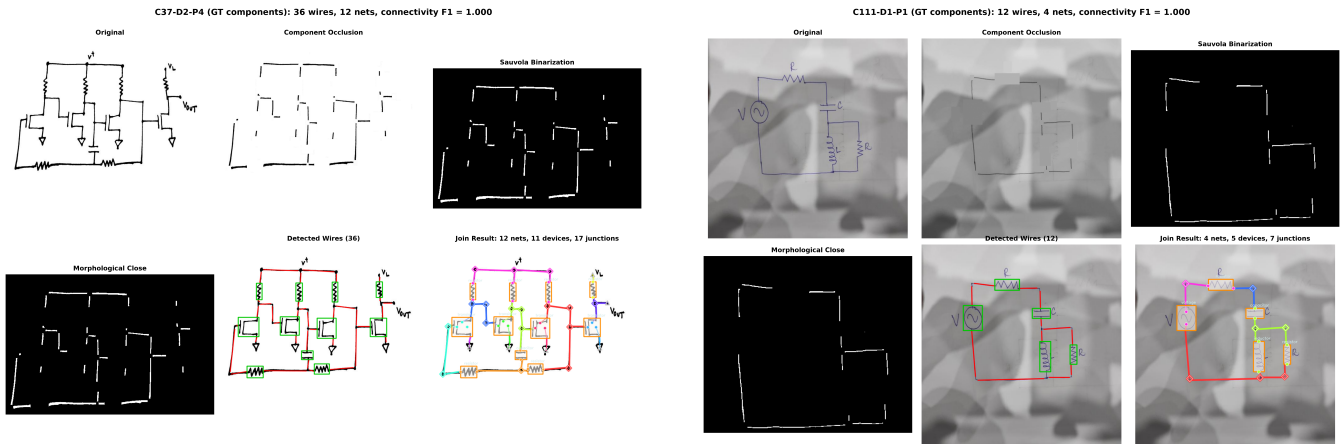


FIGURE 2. Pipeline stages on two real hand-drawn circuit images, joined with our scale-relative completion join (connectivity $F1 = 1.000$ on both). Left: a four-transistor analog cell (36 wires, 12 nets); right: an RLC loop photographed on a textured surface (12 wires, 4 nets). Each panel shows, in order: original scan, component occlusion, Sauvola binarization, morphological close, detected wires (red overlay), and the join result. In the join result every recovered net is a distinct colour drawn along its conductors; component bounding boxes are the ground-truth labels (as used throughout the net-level evaluation, isolating the join), filled dots are device pins, diamonds mark junctions, and triangles mark terminals and ground.

endpoint positions that vary by 10–80 pixels. Second, components occupy bounding boxes that overlap with wire endpoints, creating ambiguity about whether an endpoint connects to a component’s pin or merely passes nearby. Third, T-junctions, collinear fragments, and crossing wires create complex connectivity patterns that simple radius-based approaches cannot resolve without over-merging.

A natural alternative is to prompt a large vision-language model (VLM) to read the schematic and emit its netlist directly. We test this in Section IV-F: a strong VLM attains connectivity accuracy statistically indistinguishable from our geometric pipeline, but it does so at orders-of-magnitude higher cost, returns free-form text rather than a structured netlist, offers no structural guarantee against electrically invalid output (e.g. shorted two-terminal parts), and drops connections on the most complex circuits. These properties motivate an explicit, inspectable geometric pipeline whose output is a netlist by construction and whose accuracy matches the VLM’s while being deterministic and simulatable.

We address these challenges with a complete deterministic pipeline whose novel elements are: (1) an occlusion-first wire extractor that masks detected components before binarization and estimates per-segment endpoints, producing a line-segment-with-endpoints rep-

resentation (rather than tracing nets directly into connected components, as prior systems do) that is what makes graph-based joining possible; (2) an endpoint-graph join model that constructs a graph over both wire endpoints and component pins, with five edge types capturing distinct connectivity patterns; and (3) a degree-budget completion algorithm that identifies and reconnects dropped connections using constrained optimization. We evaluate on the CGHD-1152 dataset [1] with both a synthetic error injection framework that enables systematic robustness analysis and a human-verified net-level benchmark.

II. RELATED WORK

The end-to-end conversion of schematic images into machine-readable netlists has progressed rapidly. Kelly and Cole [2] presented a pattern-recognition workflow for digitizing circuit schematics with an 86.4% success rate, combining optical character recognition (OCR) for component values with network-graph construction. SINA [3], the closest contemporary system, pairs a YOLOv11 detector with connected-component labeling (CCL) and OCR, and invokes a vision-language model (VLM) for reference-designator assignment, reporting 96.47% netlist-generation accuracy, a $2.72\times$ improvement over the prior state of the art. SINA confines the

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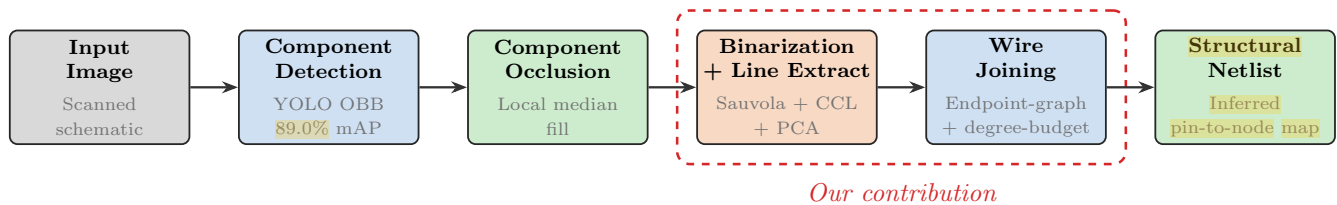


FIGURE 1. Pipeline for structural circuit-netlist recovery. The highlighted stages extract wire endpoints and infer connectivity. The net-level experiments use annotated component boxes to isolate joining; they do not establish an end-to-end error hierarchy or validate illustrative SPICE export.

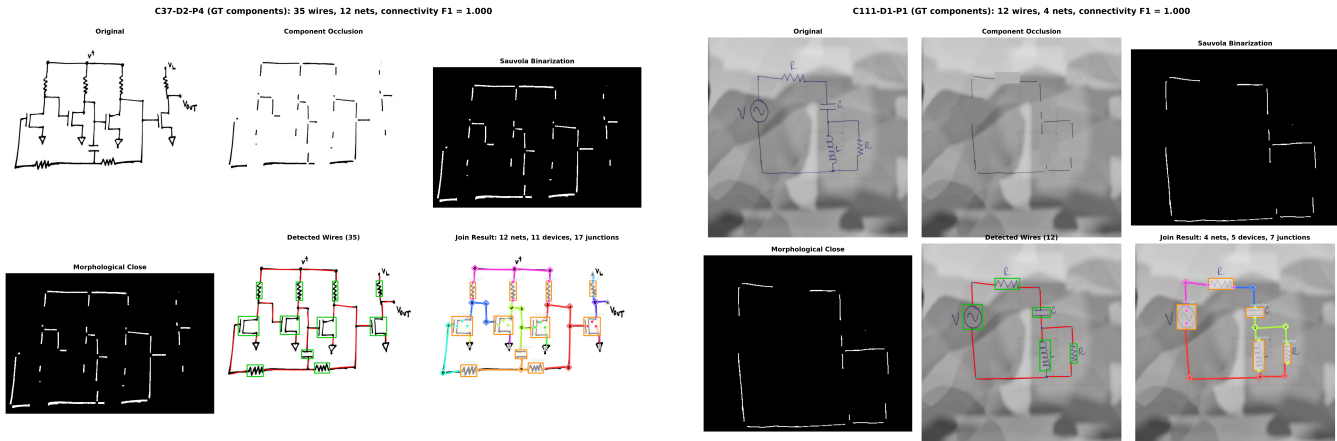


FIGURE 2. Pipeline stages on two real hand-drawn circuit images, joined with our scale-relative completion join (connectivity F1 = 1.000 on both). Left: a four-transistor analog cell (35 wires, 7 nets); right: an RLC loop photographed on a textured surface (11 wires, 4 nets). Each panel shows, in order: original scan, component occlusion, Sauvola binarization, morphological close, detected wires (red overlay), and the join result. In the join result every recovered net is a distinct colour drawn along its conductors; component bounding boxes are the ground-truth labels (as used throughout the net-level evaluation, isolating the join), filled dots are device pins, diamonds mark junctions, and triangles mark terminals and ground.

scores do not establish equivalence, comparative cost superiority or a global structural-validity guarantee. The diagnostic examines the methods under those stated input conditions.

We address these challenges with a deterministic structural-netlist pipeline whose proposed elements are: (1) an occlusion-first wire extractor that masks detected components before binarization and estimates per-segment endpoints, producing a line-segment-with-endpoints representation (rather than tracing nets directly into connected components, as prior systems do) that is what makes graph-based joining possible; (2) an endpoint-graph join model that constructs a graph over both wire endpoints and component pins, with five edge types capturing distinct connectivity patterns; and (3) a degree-budget completion algorithm that proposes recovery connections using bounded assignment and merge guards. We evaluate on the CGHD-1152 dataset [1] with a synthetic error injection framework and a human-verified net-level benchmark. We evaluate topological recovery with illustrative SPICE export; component values and device models require external specification, and simulation equivalence on real scans is not validated. The output records inferred pin-to-node assignments, whereas the primary metric measures component-pair connectivity rather than exact pin topology.

II. RELATED WORK

The end-to-end conversion of schematic images into machine-readable netlists has progressed rapidly. Kelly and Cole [2] presented a pattern-recognition workflow for digitizing circuit schematics with an 86.4% success rate, combining optical character recognition (OCR) for component values with network-graph construction. SINA [3], the closest contemporary system, pairs a YOLOv11 detector with connected-component labeling (CCL) and OCR, and invokes a vision-language model (VLM) for reference-designator assignment, reporting 96.47% netlist-generation accuracy, a $2.72\times$ improvement over the prior state of the art. SINA confines the VLM to symbolic labeling and derives nets from CCL proximity rather than trusting the VLM with connectivity, an insight we share (Section II). Masala-CHAI [4] (Auto-SPICE) uses large language models to produce SPICE netlists from analog schematics at scale. Most directly comparable to our setting, the concurrent work of Peker et al. [5] (IEEE Access, 2026) builds a fully automated image-to-SPICE pipeline for printed and hand-drawn transistor-level circuits, pairing YOLOv8/11 detection with a contour-based node-detection module (components are erased and nets are read from the contours of the residual wire pixels) and validates netlists by automated LTspice simulation against a reference. On their in-house seven-topology MOSFET dataset they

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Progress depends on annotated corpora. The CGHD hand-drawing dataset [1] provides component- and **wire-level annotations** for handwritten schematics. In the **analog/mixed-signal** (AMS) domain, AMSnet 2.0 [13] contributes a large database built with AI-assisted segmentation for net **detection**, while AMSBench [14] is a benchmark for **evaluating** multimodal LLM **capabilities** across AMS circuit tasks. Beyond circuits, DiagramNet [15] introduces an end-to-end framework and dataset for non-standard system-level **diagrams**, including an explicit connection-recognition task scored by F1. **We use these benchmarks both as evidence of where current systems succeed (component recognition) and where they fail (connectivity), and contribute a synthetic robustness protocol that isolates the join step under controlled detector noise.**

General-purpose VLMs and LLMs have been applied

broadly to electronic design [16], and to circuit **understanding** in particular. Kulkarni et al. [17] proposed Near Sight Correction using VLMs, reporting 87.4% accuracy on circuit visual question answering (VQA) and 0.87 F1 on **connection-matrix** prediction. **However, multiple independent benchmarks show that general VLMs handle component recognition far better than connectivity.** On the DiagramNet connection task [15], leading general models score very low F1 (GPT-5 at 0.029, Claude-Sonnet-4 at 0.265, and Gemini-2.5-Pro at 0.008), whereas a task-tuned 3B model reaches 0.735. AMSBench [14] reports a similar gap for MLLMs on AMS connectivity. **This corroborates the design choice, shared with SINA [3], to keep wire-to-component connectivity out of the VLM and instead solve it with an explicit geometric model. We make this separation concrete and quantify it with our own Claude-VLM connectivity experiment.**

Classical wire extraction relies on edge detection, **morphological** operations, and the Hough transform [18], typically preceded by adaptive binarization such as Sauvola thresholding [19] and connected-component labeling [20]. The dominant join in existing pipelines uses radius-based proximity: each wire endpoint grabs all component pins within a fixed threshold (typically 30px), connected via transitive union-find [21]. This **over-merges** when multiple components fall within the threshold, **shorting** two-terminal **parts** (R, C, L, D) onto the same net. SINA [3] likewise derives nets from CCL proximity, which is similarly vulnerable to **over-merge** under detector noise. We instead model joining as a typed endpoint graph that incorporates component geometry and directional **preference**, and resolve residual ambiguity with a degree-budget completion: a bounded b-matching formulation [22], [23] that caps per-pin **degree** and thereby limits over-merge when **endpoints** are dropped or displaced. This differs from graph **neural** networks for netlist reverse engineering [24], which learn representations over already-extracted gate-level graphs; our graph is a purely geometric model whose edges encode spatial relationships between wire endpoints and component pins. A complementary recent direction replaces rule-based **connectivity** with a learned model: Netlistify [25] predicts terminal connectivity with a Transformer and reports a 12.4% F1 gain over AMSnet on printed AMS schematics. Such models **are powerful but require** task-specific connectivity training data and, **trained on printed schematics, do not transfer to hand-drawn images without retraining; our geometric model needs no connectivity training and is the focus here.** Open hand-drawn efforts such as CircuitNet [26] likewise rely on traditional image processing plus a classifier and a heuristic netlist generator.

III. METHOD

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report 93.33% (printed) and 85.33% (hand-drawn) whole-netlist success. Their node detection is an instance of the connected-component/contour recipe we evaluate as a baseline (Section IV-D); because their dataset (transistor-level analog, authored in-house) and metric (LTspice node-voltage pass/fail) differ from ours (discrete components on CGHD-1152, component-pair F1), the headline numbers are not directly comparable. Our controlled comparison evaluates a CCL baseline on identical detected wires, rather than reproducing their complete system. Instance-segmentation approaches extract circuit graphs directly from pixels: Bayer et al. [8] segment components and wires for graph extraction over the CGHD lineage. For hand-drawn inputs, object-detection pipelines [9] and the recent JUHCCR-v1 database [10] target component recognition on freehand sketches. Most of these pipelines build on general-purpose object detectors such as the YOLO family [11], [12] and OCR engines such as Tesseract [13], with the ultimate goal of producing simulator-ready descriptions in formats such as SPICE [14]. These works address detection, recognition and connectivity under different datasets and metrics. Our contribution evaluates joining under specified component priors, without treating detection or recognition as solved. SPICE-level circuit modeling likewise underpins application deployment in neighboring hardware domains, including memristor-based neuromorphic circuits that realize brain-inspired learning in hardware [6] and memristive decision-making frameworks for autonomous navigation [7].

Progress depends on annotated corpora. The CGHD hand-drawing dataset [1] provides component- and wire-level annotations for handwritten schematics. In the analog/mixed-signal (AMS) domain, AMSnet 2.0 [15] contributes a large database built with AI-assisted segmentation for net detection, while AMSBench [16] is a benchmark for evaluating multimodal LLM capabilities across AMS circuit tasks. Beyond circuits, DiagramNet [17] introduces an end-to-end framework and dataset for non-standard system-level diagrams, including an explicit connection-recognition task scored by F1. These benchmarks provide context for component recognition and connectivity evaluation. Our synthetic perturbation protocol isolates joining under authored errors; it does not measure the relative autonomous error contributions of those stages.

General-purpose VLMs and LLMs have been applied broadly to electronic design [18], and to circuit understanding in particular. Kulkarni et al. [19] proposed Near Sight Correction using VLMs, reporting 87.4% accuracy on circuit visual question answering (VQA) and 0.87 F1 on connection-matrix prediction. Connectivity-specific benchmarks report results that depend on the task, model and input conditions. On the DiagramNet connection task [17], leading general models score very low F1 (GPT-5 at 0.029, Claude-Sonnet-4 at 0.265, and Gemini-2.5-Pro at 0.008), whereas a task-tuned 3B model reaches 0.735. AMSBench [16] reports a similar gap for MLLMs on AMS connectivity. These task-specific results motivate a separately scoped connectivity diagnostic;

they do not establish that geometry is superior to VLMs on our corpus. Our Claude-VLM experiment uses matched annotated component priors, as described in Section IV-H.

Classical wire extraction relies on edge detection, morphological operations, and the Hough transform [20], typically preceded by adaptive binarization such as Sauvola thresholding [21] and connected-component labeling [22]. Our legacy join baseline uses radius-based proximity: each wire endpoint grabs all component pins within a fixed threshold (typically 30px), connected via transitive union-find [23]. This can over-merge when multiple components fall within the threshold, assigning both pins of a two-terminal part (R, C, L, D) to the same net. SINA [3] likewise derives nets from CCL proximity, which is similarly vulnerable to over-merge under detector noise. We instead model joining as a typed endpoint graph that incorporates component geometry and directional preference, and resolve residual ambiguity with a degree-budget completion: a bounded b -matching formulation [24], [25] that caps per-pin degree and thereby limits over-merge when endpoints are dropped or displaced. This differs from graph neural networks for netlist reverse engineering [26], which learn representations over already-extracted gate-level graphs; our graph is a purely geometric model whose edges encode spatial relationships between wire endpoints and component pins. A complementary recent direction replaces rule-based connectivity with a learned model: Netlistify [27] predicts terminal connectivity with a Transformer and reports a 12.4% F1 gain over AMSnet on printed AMS schematics. Such models use task-specific connectivity training data. Transfer of the cited printed-schematic model to this hand-drawn benchmark is not established here; our geometric join requires no connectivity training. Open hand-drawn efforts such as CircuitNet [28] likewise rely on traditional image processing plus a classifier and a heuristic netlist generator.

III. METHOD

A. PIPELINE OVERVIEW

Figure 1 summarizes the six-stage pipeline. The full pipeline proceeds through six stages: (1) component detection via oriented bounding box (OBB) YOLO model; (2) component occlusion (filling detected regions with local median color); (3) Sauvola adaptive binarization; (4) connected component labeling with morphological post-processing; (5) wire endpoint extraction via principal component analysis (PCA) on endpoint regions; and (6) wire joining via the endpoint-graph model with degree-budget completion. The output is a structural netlist mapping inferred component pins to electrical nodes. Component-pair evaluation does not certify those pin assignments or the exact net partition. SPICE export is illustrative: component values and device models require external specification, and simulation equivalence on real scans is not validated. Figure 2 shows two representative circuits at each stage.

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Related work and framing

Review suggested references and the scope of comparisons.

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B. ENDPOINT-GRAPH JOIN MODEL

Given a set of wire segments $\mathcal{W} = \{w_1, \dots, w_m\}$ with endpoints $\text{ep}(w_i) = (a_i, b_i)$ and component pins $\mathcal{P} = \{p_1, \dots, p_n\}$, we construct an undirected graph $G = (V, E)$ where $V = \text{eps}(\mathcal{W}) \cup \mathcal{P}$.

Edge type 1 (wire body): For each wire w_i , add edge (a_i, b_i) with weight 0. This ensures both endpoints of a wire are always in the same net.

Edge type 2 (endpoint-endpoint): For endpoint pairs (e_i, e_j) with $\|e_i - e_j\| \leq \tau_{\text{join}}$, add edge with weight $\|e_i - e_j\|$. This captures wire fragments, corners, and junctions.

Edge type 3 (endpoint-pin): For each endpoint e and pin p with $\|e - p\| \leq \tau_{\text{pin}}$, add edge with weight $\|e - p\| \cdot (1 - \alpha \cdot \cos \theta)$, where θ is the angle between the wire direction and the endpoint-to-pin vector, and α controls directional preference. This is the critical edge type: it connects wires to components.

Edge type 4 (endpoint-wire-body): For each endpoint e and wire segment $w_j = (c, d)$, if the point-to-segment distance $\text{dist}(e, w_j) \leq \tau_t$, add edge with weight $\text{dist}(e, w_j)$. This captures T-junctions where a wire terminates on another wire's body.

Edge type 5 (pin-wire-body): For each component pin p and wire segment w_j , if the point-to-segment distance $\text{dist}(p, w_j) \leq \tau_t$ and the projection lands mid-span (not at w_j 's endpoints, which are handled by edge type 3), add edge with weight $\text{dist}(p, w_j)$. This binds components that tap onto a passing rail along the wire's body rather than at its end. It recovers connectivity that endpoint-only binding misses, without the all-to-all over-merge of radius grabbing.

The tolerances τ_{join} , τ_{pin} , and τ_t are scale-relative: $\tau = k \cdot s$ where s is the characteristic scale (median component diagonal), so a single parameter setting works across circuits of varying size.

Component-first pin assignment: For edge type 3, we use a two-step assignment. First, assign each endpoint to its nearest component by bounding-box distance (0 if inside the bbox, using a generous radius of $\max(\tau_{\text{pin}}, 0.5 \cdot \text{diagonal})$). Second, within the assigned

component, select the pin by geometric position: for horizontal components ($w > h$), left pin if endpoint $x < \text{center } x$, else right pin; for vertical components ($h > w$), top pin if endpoint $y < \text{center } y$, else bottom pin. This handles the common case where endpoints land inside a component's bounding box but far from its pin. Figure 3 contrasts the legacy radius join with our endpoint-graph model.

Nets are computed as connected components of G , projected onto pins. The result is a netlist mapping each pin to a net identifier.

C. DEGREE-BUDGET COMPLETION

The endpoint-graph join may leave some component pins "floating": touching only their own component's net, with no wire connecting them to another component. This indicates a dropped or over-displaced wire endpoint that the detector failed to place near the correct pin.

Degree-budget completion identifies these floating pins and reconnects them via constrained optimization. A pin p is treated as floating if its net contains only pins from p 's own component, excluding inert types such as gnd, vss, text, antenna, junction, and terminal. For each floating pin p , we generate candidate edges to all pins q in other components within reach $r = \rho \cdot s$, where ρ is the reach factor; we use $\rho = 4$ for the deployed scale-relative join, and Section IV-D shows a broad F1 plateau for $\rho \in [3, 5]$. Candidate edge weight is Euclidean distance $\|p - q\|$. We then solve a min-cost b-matching problem, $\min \sum_{(p,q) \in E'} w(p,q) \cdot x_{pq}$ subject to $\sum_q x_{pq} \leq 1$ for each floating pin p , using the Hungarian algorithm [22] (`scipy.optimize.linear_sum_assignment`). Each target pin offers up to three slots so legitimate multi-terminal nets are not starved, and a per-pin opt-out leaves a pin with no good target unconnected rather than forcing a spurious edge. Finally, we reject any matched edge (p, q) if p and q belong to the same component or if adding the edge would merge two nets that already share a component, which structurally prevents short-circuiting two-terminal parts.

Completion edges are "wireless": they represent inferred connections without corresponding wire segments, analogous to manual pin merges in schematic editors. Figure 4 illustrates the completion step.

IV. RESULTS

A. SYNTHETIC EVALUATION FRAMEWORK

We author 15 synthetic circuits spanning 3–6 components and 2–6 nets, including parallel resistors, series dividers, Wheatstone bridges, RC/RL circuits, diode loops, and a 6-component ring (cyclic and bridge topologies being the main stress cases for join robustness). Each circuit has a known ground-truth netlist and SPICE deck with a test voltage source.

We inject five categories of error at four severity levels (L1–L4) on top of the clean (L0) baseline. These

ROSE = removed / replaced text

How to use this pair

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Baseline: pre-August-13 Git snapshot 33f5e3d; not independently portal-confirmed.

Write the contributor and decision on the following NEW page.

PROBLEMS / NO PROBLEMS

LIMITATION / CHECK

Device-specific pins are not fully validated.

SUGGESTED ACTION

Keep generic geometry distinct from verified device behavior; check the response wording.

Attribution note: Pranavesh authored earlier benchmark commits. Git lists Bosco Chanam as author of the later manuscript revision commits; confirm individual edit credit with the authors.

B. ENDPOINT-GRAPH JOIN MODEL

Given a set of wire segments $\mathcal{W} = \{w_1, \dots, w_m\}$ with endpoints $\text{ep}(w_i) = (a_i, b_i)$ and component pins $\mathcal{P} = \{p_1, \dots, p_n\}$, we construct an undirected graph $G = (V, E)$ where $V = \text{eps}(\mathcal{W}) \cup \mathcal{P}$.

Edge type 1 (wire body): For each wire w_i , add edge (a_i, b_i) so both endpoints belong to the same net. The implementation accumulates connectivity with union-find; it does not solve a weighted graph optimization.

Edge type 2 (endpoint–endpoint): For endpoints e_i, e_j on different wires with $\|e_i - e_j\| \leq \tau_{\text{join}}$, add an edge. This proximity rule can connect fragments, corners and junctions, but can also merge unrelated nearby wires.

Edge type 3 (endpoint–pin): Each endpoint is connected to at most one selected pin by this edge rule. First apply the component-first assignment described below. Only if no component is assigned, search pins within Euclidean distance $d \leq \tau_{\text{pin}}$ and select the pin with the smallest score. The score is d unless directional preference is enabled and $d > 10^{-6}$ px; in that case it is $d[1 - \alpha \max(0, \cos \theta)]$, with fixed $\alpha = 0.35$ and θ measured from the outward wire direction to the endpoint-to-pin vector. Directional preference is enabled in the scale-relative base, but applies only to this fallback search. If an assigned component has no eligible pin, or the fallback finds none, no endpoint–pin edge is added. The score selects a fallback pin; it is not a weight optimized over the final graph.

Edge type 4 (endpoint–wire-body): For an endpoint e and a different wire segment w_j , add an edge if the point-to-segment distance is at most τ_t and the normalized projection parameter lies in $[0.15, 0.85]$. This rule represents T-junction candidates away from the segment ends.

Edge type 5 (pin–wire-body): For a component pin p and wire segment w_j , add an edge if the point-to-segment distance is at most τ_t and the normalized projection parameter lies in $[0.05, 0.95]$. This rule represents pin-to-rail candidates along a wire’s body; geometric proximity alone does not certify an electrical connection.

The tolerances are manually configured scale-relative rules with fixed pixel clamps: $\tau_{\text{pin}} = \min(60, \max(24, 0.62s))$, $\tau_{\text{join}} = \min(28, \max(11, 0.30s))$, and $\tau_t = \min(20, \max(8, 0.20s))$ in pixels. Here s is the characteristic component-bounding-box diagonal scale; the implementation uses the middle sorted diagonal (the upper middle for an even count), falling back to a wire-length scale when component geometry is unavailable or degenerate. The multipliers and clamps are fixed, not learned. Because s is a single scalar per image, intra-image size variance is not modeled: a schematic mixing a large IC with small discrete symbols shares one tolerance per rule. Extreme-size performance was not evaluated by controlled image rescaling.

Component-first pin assignment: For edge type 3, we use a two-step assignment. First, compute the distance from each endpoint to each component’s oriented bounding box (OBB): 0 inside the rectangle, otherwise the minimum distance to its edges. If four OBB vertices are unavailable, use axis-aligned bounding-box (AABB) distance instead. Assign the

endpoint to the nearest component whose distance is within $\max(\tau_{\text{pin}}, 0.5 \cdot \text{diagonal})$, where the diagonal is computed from that component’s AABB. Second, within the assigned component, select the pin with the smallest Euclidean distance to the endpoint. Figure 3 contrasts the legacy radius join with our endpoint-graph model.

Nets are computed as connected components of G , projected onto pins. The result is a netlist mapping each pin to a net identifier.

C. DEGREE-BUDGET COMPLETION

The endpoint-graph join may leave a pin in a net containing only pins from its own component. Completion treats such pins as candidates for recovery, excluding inert types such as ground, supply markers, text, antenna, probes, junctions and terminals. This condition can reflect a missing connection, but does not establish that the detector dropped a wire.

Degree-budget completion retains the base net assignments and constructs candidates from each floating pin p to pins q on other components and different current nets. Candidate distance must not exceed $r = \rho \tau_c$, where $\tau_c = \min(60, \max(24, 0.62s))$ px and the deployed scale-relative completion uses $\rho = 4$. Thus reach uses the clamped pin scale, not the raw component diagonal; completion retains this scale dependence when base tolerances are ablated. For each candidate, a wire witness is sought by pairing the endpoints of a detected segment with p, q in either orientation. The witness cost is the smallest, over feasible segments and orientations, of the larger endpoint-to-pin distance, bounded by r . If no witness exists, the deployed relaxed-witness mode uses $\|p - q\|$ as the candidate cost.

When candidates exist, the implementation builds a rectangular assignment matrix with one row per floating pin. Each target pin supplies up to three distinct slot columns, limited by its candidate demand, and each row has a private dummy column with cost $1.5r$ for opting out. Unavailable assignments carry sentinel cost 10^6 . The Hungarian algorithm [24] (`scipy.optimize.linear_sum_assignment`) assigns every row to exactly one column, with at most one row per column. This mandatory assignment with dummy options avoids the empty-solution ambiguity of a nonnegative-cost objective with only at-most-one constraints; target slots implement the bounded-capacity matching [25].

Matches are then applied sequentially. Dummy or unavailable assignments add no edge. A proposed merge is also skipped if its endpoints are already in the same net or if their current nets share a component. This guard limits additional completion merges; it does not undo incorrect base unions or guarantee globally short-free output. Rejected matches are not reassigned, so the final guarded result is not claimed to minimize the assignment objective over all valid netlists.

Completion edges are “wireless”: they represent inferred connections without corresponding wire segments, analogous to manual pin merges in schematic editors. Figure 4 illustrates the completion step.

GOLD = added / revised text

Method and endpoint graph

Methods and implemented pin/join rules have been reconciled.

OPEN: device-specific pin behavior is not fully validated.

PROBLEMS / NO PROBLEMS**LIMITATION / CHECK**

Device-specific pins are not fully validated.

SUGGESTED ACTION

Keep generic geometry distinct from verified device behavior; check the response wording.

YOUR REVIEW STATUS**CHANGE OWNER / CONTRIBUTOR****COMMENTS / DECISION**

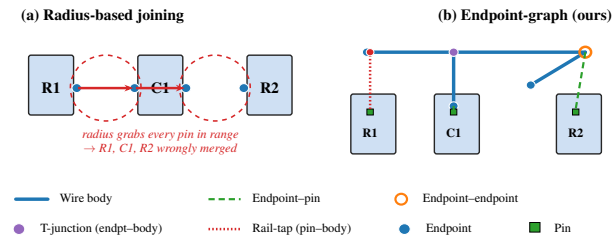


FIGURE 3. Joining approaches compared. (a) Radius-based join (legacy): each wire endpoint grabs all pins within a fixed threshold, causing over-merging. (b) Our endpoint-graph model: typed edges (wire body, endpoint-pin) with directional preference resolve connectivity without over-merging.

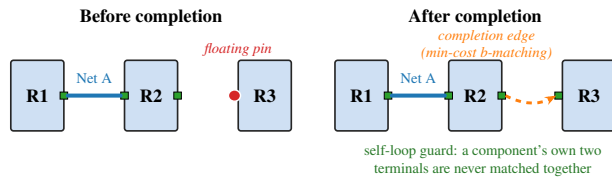


FIGURE 4. Degree-budget completion. Left: After base joining, some pins are “floating” (touching only their own component). Right: Completion reconnects floating pins via min-cost b -matching (orange dashed), with self-loop guard preventing shorts within two-terminal components.

are Gaussian endpoint jitter with standard deviation σ (from 3 px at L1 to 14 px at L4), cut-short corruption in which each wire is shortened inward by a fixed length to model endpoints that stop short of the pin (from 4 px at L1 to 22 px at L4), full wire drops with probability p_{drop} (up to 0.20 at L4), wrong-pin snaps with probability p_{wp} (from 0.02 at L2 to 0.10 at L4), and anchor displacement, where with probability p_{disp} one endpoint is moved by up to a_{max} pixels (from 30 px at L1 to 80 px at L4).

We report join F1 as the component-connectivity F1 score, measuring which pairs of components share a net in the predicted versus ground-truth netlist. We report simulation accuracy as the percentage of random seeds for which every source current in the predicted SPICE netlist matches the clean oracle within 5%.

B. WIRE DETECTION BENCHMARK

We evaluate on 134 images from CGHD-1152 with ground-truth wire and component labels. Figure 5 and Table 1 summarize the top configurations across 36 tested variants.

TABLE 1. Wire Detection Performance (134 images)

Configuration	F1	Prec	Rec
Sauvola + anchor 16 px (ours)	0.976	0.973	0.978
Sauvola + anchor 12 px	0.973	0.974	0.972
Sauvola ($w=61$) + anchor 12 px	0.959	0.944	0.974
Sauvola ($k=0.2875$) + anchor 14 px	0.950	0.921	0.980
Otsu threshold + CCL	0.789	0.796	0.783

The pipeline achieves median F1 = 1.000 across the benchmark set, with 87% of images scoring F1 $\geq$ 0.90. Sauvola adaptive binarization dominates all alternatives: Otsu yields F1 = 0.789, adaptive Gaussian

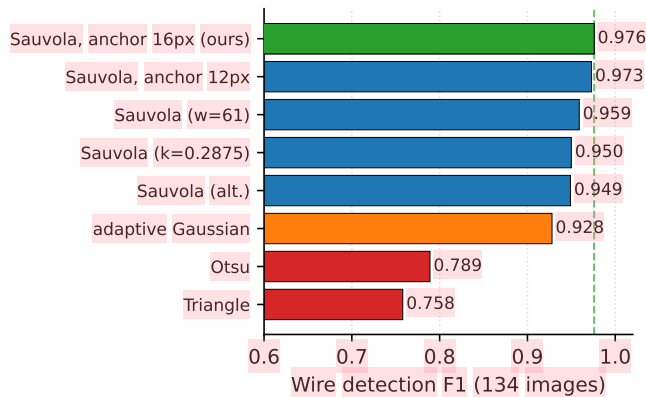


FIGURE 5. Wire detection F1 across thresholding methods and pipeline variants (134 images). Our configuration (Sauvola binarization with a 16 px anchor filter) achieves F1 = 0.976. Otsu and triangle thresholding degrade significantly.

F1 = 0.845. Component occlusion matters: without it, component internals (text, symbols, internal wiring) generate hundreds of false wire detections.

C. JOIN STRATEGY COMPARISON

Table 2 and Figure 6 compare joining strategies on the synthetic evaluation framework (15 circuits $\times$ 8 seeds, mean F1 over error levels L1–L4).

The completion-based strategies are the most robust, maintaining F1 $\geq$ 0.94 at the highest error severity (L4), compared to F1 = 0.36 for the legacy radius union-find baseline, a 2.6 $\times$ improvement. The radius join’s transitive grabbing causes catastrophic over-merging under noise: at L4, it shorts multiple components onto shared nets. We emphasize, however, that the endpoint-graph base joins are themselves far more robust than the radius join, reaching F1 = 0.90 at L4 (Table 2). The

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Baseline: pre-August-13 Git snapshot 33f5e3d; not independently portal-confirmed.

Write the contributor and decision on the following NEW page.

PROBLEMS / NO PROBLEMS

OPEN EVIDENCE GAP

Wire rerun used 704x704 identity copies, not full-resolution CGHD scans.

SUGGESTED ACTION

Name the preprocessed image domain; do not claim original-resolution performance.

Attribution note: Pranavesh authored earlier benchmark commits. Git lists Bosco Chanam as author of the later manuscript revision commits; confirm individual edit credit with the authors.

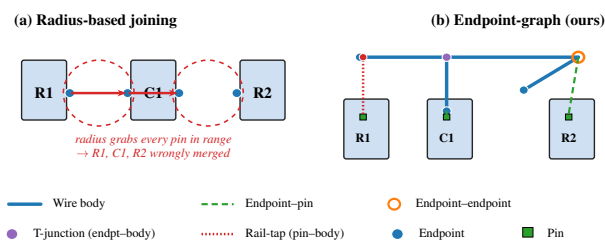


FIGURE 3. Joining approaches compared. (a) Radius-based join (legacy): each wire endpoint grabs all pins within a fixed threshold, causing over-merging. (b) Our endpoint-graph model: typed edges and component-first pin assignment generate connectivity candidates, with directional scoring confined to the fallback pin search. Incorrect base merges remain possible.

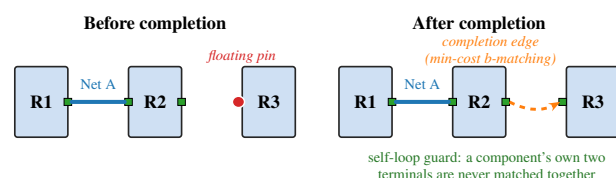


FIGURE 4. Degree-budget completion. Left: After base joining, some pins are “floating” (touching only their own component). Right: Completion reconnects floating pins via min-cost b -matching (orange dashed), with a guard rejecting additional merges between nets that already share a component; existing base shorts are not repaired.

IV. RESULTS

A. SYNTHETIC EVALUATION FRAMEWORK

We author 15 synthetic circuits spanning 3–6 components and 2–6 nets, including parallel resistors, series dividers, Wheatstone bridges, RC/RL circuits, diode loops, and a 6-component ring (cyclic and bridge topologies being the main stress cases for join robustness). Each circuit has a known ground-truth netlist and SPICE deck with a test voltage source.

We inject five categories of error at four severity levels (L1–L4) on top of the clean (L0) baseline. These are Gaussian endpoint jitter with standard deviation σ (from 3 px at L1 to 14 px at L4), cut-short corruption in which each wire is shortened inward by a fixed length to model endpoints that stop short of the pin (from 4 px at L1 to 22 px at L4), full wire drops with probability p_{drop} (up to 0.20 at L4), wrong-pin snaps with probability p_{wp} (from 0.02 at L2 to 0.10 at L4), and anchor displacement, where with probability p_{disp} one endpoint is moved by up to a_{max} pixels (from 30 px at L1 to 80 px at L4).

We report join F1 as the component-connectivity F1 score, measuring which unordered pairs of components share a net in the predicted versus ground-truth netlist. This metric discards terminal identity; even a perfect component-pair score does not certify pin assignment, the exact net partition, absence of shorts, or simulation equivalence. We report simulation accuracy as the percentage of random seeds for which every source current in the predicted SPICE netlist matches the clean oracle within 5%.

B. WIRE DETECTION BENCHMARK

We evaluate on 134 images from CGHD-1152 with ground-truth wire and component labels. Figure 5 and Table 1 sum-

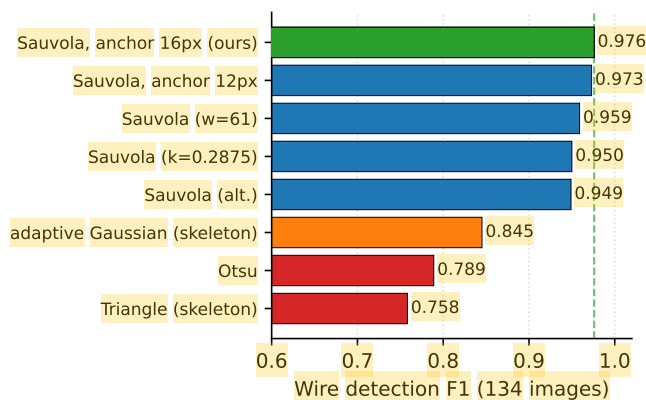


FIGURE 5. Wire detection F1 across the tested thresholding methods and pipeline variants (134 images, annotated component boxes). The historical Sauvola configuration with a 16 px anchor filter and overlap deduplication at $10^{\circ}/18$ px achieves $F1 = 0.976$; this is distinct from the production $12^{\circ}/8$ px deduplication setting. The adaptive Gaussian and Triangle bars use the stored skeleton-extraction configurations ($F1 = 0.845$ and 0.758 , respectively).

marize the top configurations across 36 tested variants.

The historical 16 px-anchor configuration achieves median F1 1.000, with 87% of benchmark images scoring F1 at least 0.90. Its aggregate F1 is 0.9755 at $10^\circ/18$ px overlap deduplication; the $12^\circ/8$ px production configuration gives 0.9726 with the same 16 px anchor distance. Among the tested alternatives, Otsu yields F1 0.789 and adaptive Gaussian 0.845. Occlusion is intended to remove component internals before wire extraction. Disabling that masking step while holding the rest of the extraction path fixed has a large measured downstream effect on the 31-image connectivity benchmark (Section IV-E): component-pair micro-F1 falls from 0.890 to 0.336. This is a mechanism ablation scored on downstream

GOLD = added / revised text

Wire benchmark and examples

134-image result reproduced on 704x704
identity copies.

OPEN: full-resolution original scans were not evaluated; label coordinate transfer is not established.

PROBLEMS / NO PROBLEMS

OPEN EVIDENCE GAP

Wire rerun used 704x704 identity copies, not full-resolution CGHD scans.

SUGGESTED ACTION

Name the preprocessed image domain; do not claim original-resolution performance.

YOUR REVIEW STATUS

CHANGE OWNER / CONTRIBUTOR

COMMENTS / DECISION

TABLE 2. Join Strategy Leaderboard (Synthetic Eval)

Strategy	L0	L1	L2	L3	L4
Scale-rel. graph + completion (ours)	1.00	1.00	1.00	0.96	0.95
Rescue graph + completion	1.00	1.00	1.00	0.95	0.94
Rescue graph (base)	1.00	1.00	1.00	0.94	0.90
Scale-rel. graph (base)	1.00	1.00	1.00	0.94	0.85
Radius union-find (legacy)	1.00	0.97	0.90	0.56	0.36
Nearest-pin + bbox anchor	1.00	0.97	0.89	0.38	0.32

degree-budget completion adds a further +0.05 over the strongest of these by recovering dropped connections the base graph join leaves floating, while its per-pin edge budget keeps precision high. Pairing that completion with the high-precision scale-relative base (the scale-relative completion join used throughout) gives the best synthetic robustness (0.95 at L4) and, more importantly, the best real-image connectivity (Section IV-D). The large gap is against the legacy radius join; against the strongest prior graph join the improvement is smaller but consistent across every severity level. For readability we omit two intermediate variants whose synthetic scores fall within rounding of rows shown (nearest-pin without anchoring, and the scale-relative base with end-extension but no dead-end rescue).

D. REAL-IMAGE NET-LEVEL EVALUATION

The synthetic results above measure robustness under a parametric error model. To validate on real detector output, we constructed, to our knowledge, the first human-verified net-level ground truth for hand-drawn circuits: 31 images from CGHD-1152 in which every electrical net was traced and confirmed by a human annotator through a purpose-built verification tool. Because one of our comparison points is a vision-language model (Section IV-F), the human, not any model, is the verifier of record, so the answer key is independent of every method under test. Net-level proposals were bootstrapped by a conservative endpoint-graph join over the dataset’s perfect wire labels and then corrected by hand; during verification the human consistently caught cases where the bootstrap join had wrongly split a single rail, and we additionally re-validate all conclusions on the independently authored synthetic ground truth to rule out residual bootstrap bias. Connectivity is scored as component-pair F1 restricted to SPICE-active components, identical to the synthetic metric.

Table 3 reports every registered join strategy on the verified set, run on the pipeline’s own detected wires. We report micro-averaged (pair-pooled) component-pair F1 as the primary metric, so that a dense 14-component board contributes in proportion to its connections rather than counting the same as a 3-component one; the macro (mean per-image) F1 is listed alongside for refer-

ence. The best configuration, the scale-relative completion join (degree-budget completion applied to a high-precision scale-relative endpoint-graph base), achieves micro-F1 = 0.890 (P = 0.92, R = 0.86; 95% bootstrap CI [0.855, 0.924]), improving on the prior default that applies the same completion to the rescue base (0.829) and the radius union-find join (0.667) (Figure 7). The gain over that rescue-based completion comes from the cleaner base: the rescue base’s 12px end-extension and dead-end rescue over-reach on noisy detected wires and cost precision, whereas the scale-relative base keeps precision high while the bounded completion still recovers recall. A reach sweep shows a broad F1 plateau across reach 3–5 × s, so the setting is robust rather than tuned to a spike. Confidence intervals are wide at N = 31 and the per-strategy intervals overlap, but the ranking is consistent and is corroborated on the independently authored synthetic suite (Table 2).

TABLE 3. Join Strategy and Baseline Comparison on Human-Verified Net-Level Ground Truth (31 real images, detected wires). Primary metric is micro-averaged (pair-pooled) component-pair F1; macro (mean per-image) F1 is shown for reference.

Method	F1	Prec	Rec	F1 _{macro}
Deterministic joins (ours and ablations)				
Scale-rel. graph + completion (ours)	0.890	0.919	0.864	0.901
Rescue graph + completion (prior)	0.829	0.789	0.874	0.853
Scale-rel. graph (base)	0.816	0.929	0.727	0.838
Rescue graph (base)	0.787	0.811	0.764	0.813
Radius union-find (legacy)	0.667	0.698	0.638	0.674
Classical deterministic baselines (identical detected wires)				
Hough + proximity (best)	0.805	0.750	0.868	0.847
Connected-components (best)	0.624	0.965	0.461	0.611
Reference (non-deterministic / oracle wires)				
VLM (Claude Opus 4.8)	0.923	0.970	0.880	0.949
Scale-rel. + completion, perfect wires	0.890	0.913	0.868	0.916

This real-image ranking resolves an open question from prior work on this pipeline: against bootstrapped pilot labels the degree-budget completion had not consistently beaten its base join; against human-verified ground truth it clearly does (micro-F1 0.829 vs 0.787 for the rescue base alone), and the scale-relative variant is better still. The same ordering holds on the authored synthetic suite (Table 2), where the scale-relative completion join also ranks first, so the result is not an artifact of how the net-level labels were bootstrapped.

The dominant connectivity recipe in prior image-to-netlist systems erases component boxes and reads nets off the connected components, or contours, of the residual wire pixels. It is used by SINA [3], the concurrent contour-based node detector of Peker et al. [5], AMSnet [13], and the Bayer instance-segmentation pipeline [6], and was designed for clean printed schematics. On hand-drawn photographs it has no precise operating point: pencil-stroke fragmentation splits each

ROSE = removed / replaced text

How to use this pair

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Pairing is by PDF page index, not exact text or section alignment.

Baseline: pre-August-13 Git snapshot 33f5e3d; not independently portal-confirmed.

Write the contributor and decision on the following NEW page.

PROBLEMS / NO PROBLEMS

OPEN HISTORICAL DISCREPANCY

Some older Pranavesh-era threshold/topology rows did not reproduce exactly.

SUGGESTED ACTION

Do not reuse old numbers as current results; reconcile any that remain in active text.

Attribution note: Pranavesh authored earlier benchmark commits. Git lists Bosco Chanam as author of the later manuscript revision commits; confirm individual edit credit with the authors.

TABLE 1. Wire detection performance on 134 images with annotated component boxes. The two Sauvola anchor-distance rows use historical 10°/18 px overlap deduplication; they are not the production 12°/8 px configuration.

Configuration	F1	Prec	Rec
Sauvola + anchor 16 px (ours)	0.976	0.973	0.978
Sauvola + anchor 12 px	0.973	0.974	0.972
Sauvola ($w=61$) + anchor 12 px	0.959	0.944	0.974
Sauvola ($k=0.2875$) + anchor 14 px	0.950	0.921	0.980
Otsu threshold + CCL	0.789	0.796	0.783

connectivity; it is not a wire-level F1 measurement for the extractor, whose human annotations are attached to the unmasked images.

C. JOIN STRATEGY COMPARISON

Table 2 and Figure 6 compare joining strategies on the synthetic evaluation framework (15 circuits $\times$ 8 seeds, mean F1 over error levels L1–L4).

TABLE 2. Join Strategy Leaderboard (Synthetic Eval)

Strategy	L0	L1	L2	L3	L4
Scale-rel. graph + completion (ours)	1.00	1.00	1.00	0.96	0.95
Rescue graph + completion	1.00	1.00	1.00	0.95	0.94
Rescue graph (base)	1.00	1.00	1.00	0.94	0.90
Scale-rel. graph (base)	1.00	1.00	1.00	0.94	0.85
Radius union-find (legacy)	1.00	0.97	0.90	0.56	0.36
Nearest-pin + bbox anchor	1.00	0.97	0.89	0.38	0.32

On the authored synthetic suite at L4, the scale-relative graph with completion scores F1 0.95, rescue graph with completion 0.94, rescue graph alone 0.90, scale-relative graph alone 0.85 and radius union-find 0.36 (Table 2). These results compare the reported configurations under the specified perturbations. They do not establish that completion repairs every dropped connection or prevents all shorts. The scale-relative completion configuration also has the highest component-pair score among the deterministic configurations reported on the real-image benchmark (Section IV-D). For readability, two intermediate synthetic variants whose scores round to rows shown are omitted: nearest-pin without anchoring, and the scale-relative base with end-extension but no dead-end rescue.

D. REAL-IMAGE NET-LEVEL EVALUATION

The synthetic results above measure robustness under an authored error model. For the real-image evaluation, we constructed a human-verified net-level benchmark of 31 CGHD-1152 images using a purpose-built verification tool. Thirty-four images were annotated; three were excluded before scoring (one pending human verification, two failing the documented physical-sanity checks), leaving the scored $N=31$. Net-level proposals were bootstrapped by a conservative endpoint-graph join over the dataset’s annotated wires and

then corrected by one human annotator. Human correction and evaluation on independently authored synthetic ground truth mitigate concern about method-dependent labels, but do not rule out residual bootstrap or annotator bias in the real benchmark. Both the geometric join and the VLM diagnostic receive annotated component boxes. Connectivity is scored as component-pair F1 restricted to the evaluated electrical-component subset, discarding terminal identities as in the synthetic connectivity metric.

Table 5 reports the evaluated join configurations on the verified set, run on the pipeline’s own detected wires. We report micro-averaged (pair-pooled) component-pair F1 as the primary metric, so that a dense 14-component board contributes in proportion to its connections rather than counting the same as a 3-component one; the macro (mean per-image) F1 is listed alongside for reference. The best configuration, the scale-relative completion join (degree-budget completion applied to a high-precision scale-relative endpoint-graph base), achieves micro-F1 = 0.890 ($P = 0.92$, $R = 0.86$; 95% bootstrap CI [0.855, 0.924]), improving on the prior default that applies the same completion to the rescue base (0.829) and the radius union-find join (0.667) (Figure 7). The scale-relative completion configuration omits the rescue base’s 12 px end-extension and dead-end rescue and has higher reported precision. This comparison does not isolate the causal contribution of those individual changes. A bounded within-benchmark reach sweep over $\rho \in [3, 5]$, with $r = \rho\tau_c$, yields macro-F1 0.895–0.903. This is sensitivity evidence for the tested settings, not a controlled image-rescaling or mixed-size robustness experiment. Confidence intervals are wide at $N = 31$ and the per-strategy intervals overlap, but the ranking is consistent and is corroborated on the independently authored synthetic suite (Table 2).

TABLE 3. Leave-one-out ablation of endpoint-graph rules at the base-graph level and with completion, using annotated component boxes and detected wires on the 31-image benchmark. All full-pipeline rows have the same aggregate TP/FP/FN (418/37/66). Fixed-pixel base tolerances give micro-F1 0.820 versus 0.816 for the scale-relative base; completion retains its scale dependence. Aggregate ties do not establish identical recovered connections or a causal explanation for the ties.

Configuration	Base graph only		
	micro-F1	TP/FP/FN	mi
Baseline (all edge types, scale-relative)	0.816	352/27/132	(
T-junctions disabled	0.816	352/27/132	(
Rail taps disabled	0.816	352/27/132	(
T-junctions + rail taps disabled	0.816	352/27/132	(
Directional preference disabled	0.816	352/27/132	(
Fixed-pixel base tolerances; completion unchanged	0.820	356/28/128	(

Table 3 reports bounded interventions on the base graph and the resulting pipeline with completion. Disabling T-junctions, rail taps or directional preference leaves the displayed base and full aggregate counts unchanged. Fixed-pixel base tolerances increase base micro-F1 from 0.816 to 0.820; the full-pipeline counts again tie at 418/37/66. Equal aggregate counts do not establish identical recovered connections, explain which mechanisms were active, or prove that completion masks differences. The baseline comparison shows an

GOLD = added / revised text

Wire and synthetic tables

36-config ranking and a16 replay match saved artifacts.

OPEN: older Pranavesh-era threshold/table rows do not all reproduce; do not silently restate them.

PROBLEMS / NO PROBLEMS

OPEN HISTORICAL DISCREPANCY

Some older Pranavesh-era threshold/topology rows did not reproduce exactly.

SUGGESTED ACTION

Do not reuse old numbers as current results; reconcile any that remain in active text.

YOUR REVIEW STATUS

CHANGE OWNER / CONTRIBUTOR

COMMENTS / DECISION

Editable fields + printable space

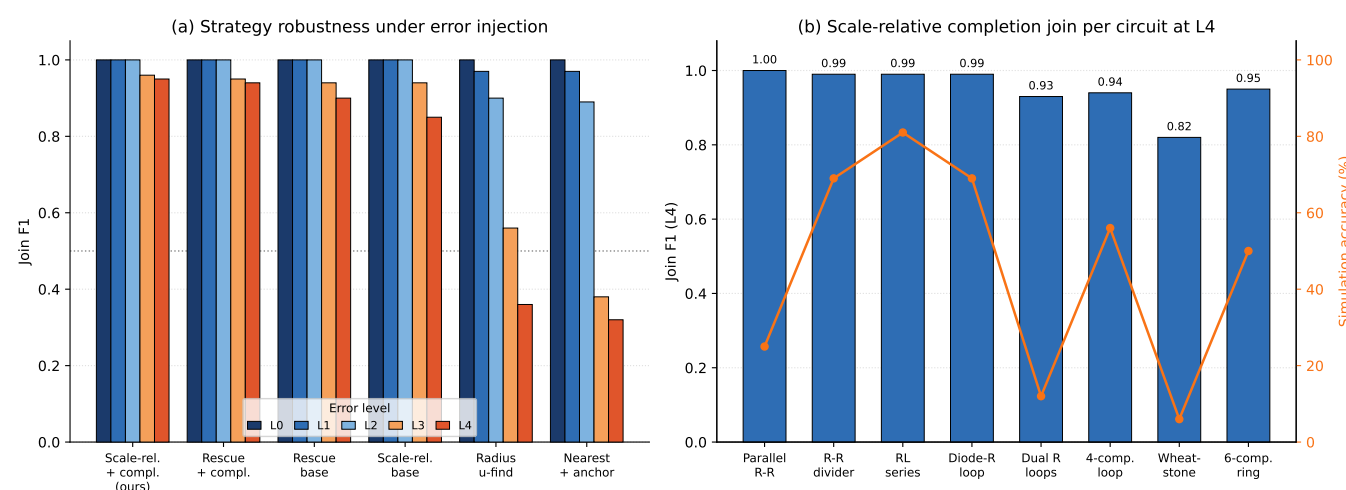


FIGURE 6. (a) Join strategy robustness under error injection (15 circuits $\times$ 8 seeds): bars give mean connectivity F1 at each severity level (L0–L4, dark to light to red). The completion-based strategies (left) stay near 1.0, while the radius union-find and nearest-pin baselines (right) collapse at L3–L4. (b) The scale-relative completion join per circuit at L4 (16 seeds): bars are connectivity F1, the orange line is simulation accuracy. Errors cause fragmentation rather than shorts (precision stays high; Table 4); the balanced Wheatstone bridge is the hardest case.

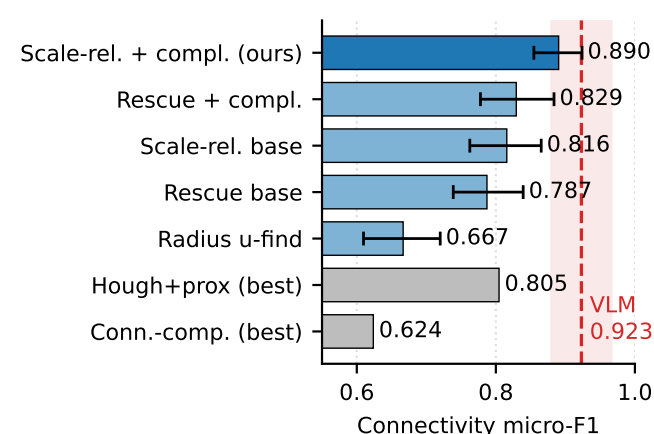


FIGURE 7. Micro-averaged connectivity F1 on the 31-image human-verified net-level benchmark (component-pair metric). Our scale-relative completion join leads all deterministic strategies and the classical Hough and connected-component baselines (run on identical detected wires). The dashed line marks a strong VLM (Claude Opus 4.8) evaluated on the same 31 images, from which our join is statistically indistinguishable (paired micro-F1 difference $+0.033$, 95% bootstrap CI $[-0.009, +0.078]$) while being deterministic, simulatable, and orders of magnitude cheaper. Error bars are 95% bootstrap CIs.

net into many components ($F1 < 0.25$ at sensible morphological gap-bridging), and only pathological closing that effectively merges by proximity lifts recall, at a steep precision cost. For a controlled comparison that isolates the join algorithm from binarization quality, we run connected-component labeling on our own detected wire segments, identical input to our join, and its best configuration reaches only micro-F1 = 0.624 (recall collapses to 0.46: the blobs short or drop most nets), versus 0.890 for the endpoint-graph join (Table 3). A stronger classical alternative, Hough-transform line detection [18] followed by proximity linking, fares better, since line fitting denoises the fragmented strokes, and

reaches micro-F1 = 0.805 at its best-tuned linking tolerance, but still trails our join by 0.085 at markedly lower precision. Each baseline is reported at its best configuration over a tolerance sweep, so these are upper bounds on the classical recipes. Explicit typed-endpoint reasoning with bounded completion thus outperforms both blob/contour connectivity and line-primitive proximity tracing.

We additionally attempted to run five public image-to-netlist systems end-to-end on our hand-drawn benchmark, and none could be fairly applied. The classical printed-schematic interpreter of Kelly and Cole [2] installs and runs, but on hand-drawn scans it over-segments wires into spurious components and recovers almost no connectivity, a domain-transfer failure. Netlistify [25] reproduces on its own printed AMS test data, which we ran end-to-end with the released model, but its active model lacks the diode, inductor, and IC classes our circuits contain and its connectivity model is trained on printed thin-line appearance, so it cannot represent hand-drawn discrete circuits. Masala-CHAI [4] delegates connectivity to a proprietary LLM behind a paid API, equivalent in spirit to our VLM experiment in Section IV-F, rather than a deterministic algorithm. SINA’s [3] anonymized code archive was unextractable and CircuitNet [26] ships only Colab notebooks without released weights. This survey is itself informative: no prior system provides a deterministic, hand-drawn-ready connectivity module, so the controlled CCL and Hough baselines above, both run on our identical detected wires, are the fair points of comparison, and our join exceeds both.

Running the best join on the dataset’s perfect wire labels rather than detected wires leaves micro-F1 unchanged at 0.890, while raising macro-F1 only from 0.901

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Write the contributor and decision on the following NEW page.

PROBLEMS / NO PROBLEMS

NO RERUN MISMATCH FOUND

The 31-image annotated-box join output matched saved per-image counts; autonomous accuracy was not shown.

SUGGESTED ACTION

Retain the annotated-box caveat and avoid end-to-end claims.

Attribution note: Pranavesh authored earlier benchmark commits. Git lists Bosco Chanam as author of the later manuscript revision commits; confirm individual edit credit with the authors.

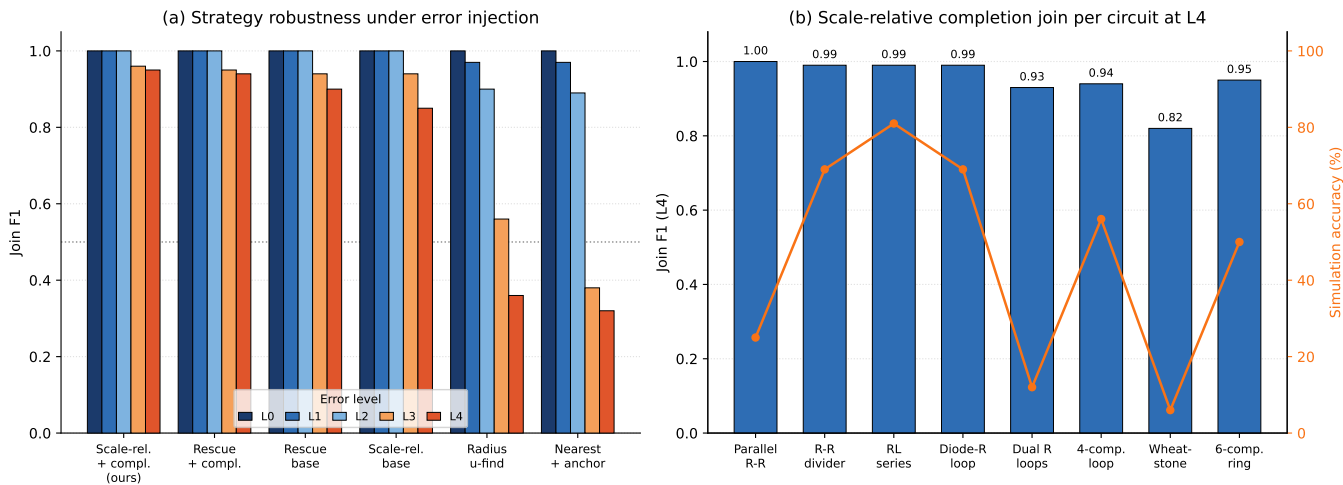


FIGURE 6. (a) Join strategy robustness under error injection (15 circuits $\times$ 8 seeds): bars give mean connectivity F1 at each severity level (L0–L4, dark to light to red). The completion-based strategies (left) stay near 1.0, while the radius union-find and nearest-pin baselines (right) collapse at L3–L4. (b) The scale-relative completion join per circuit at L4 (16 seeds): bars are connectivity F1, the orange line is simulation accuracy. **High precision does not exclude false merges or certify pin correctness; the Wheatstone bridge has the lowest reported connectivity F1 and simulation accuracy (Table 7).**

increase from base micro-F1 0.816 to 0.890 with completion, but these ablations do not establish that other mechanisms are generally unnecessary. Full occlusion removal and the completion safeguards are tested separately (Table 4, Section IV-E). Removal of scale dependence throughout completion, and extreme-size or mixed-size input conditions, were not tested. The ablation evidence therefore still does not cover every proposed module or every input condition.

E. OCCLUSION REMOVAL AND COMPLETION-SAFEGUARD ABLATIONS

Table 4 reports the mechanism interventions that Table 3 does not cover, evaluated on the same 31-image benchmark with annotated component boxes and detected wires.

TABLE 4. Mechanism ablations on the 31-image benchmark. The occlusion arm removes only the component-masking call from the extraction path; ROI cropping, extractor configuration and coordinate restoration are held fixed. Guard and witness arms act on scale_completion. “Images changed” counts images whose predicted component pairs differ from the reference arm.

Arm	TP/FP/FN	micro-F1	macro-F1	Images changed
Occlusion on (reference)	418/37/66	0.890	0.901	—
Occlusion off	103/26/381	0.336	0.294	31/31
Guard on (reference)	418/37/66	0.890	0.901	—
Guard off	418/37/66	0.890	0.901	0/31
Witness required	413/37/71	0.884	0.907	6/31

Removing component occlusion leaves component coordinates available for region-of-interest selection and anchor filtering but exposes the component pixels to the extractor, which then returns 466 wire segments instead of 666. Component-pair micro-F1 falls from 0.890 to 0.336 (precision 0.798, recall 0.213) and predicted pairs change on all 31 images. This establishes a large measured downstream contribution for the occlusion step on this benchmark. It is a mechanism ablation, not a wire-level extractor score: no

validated one-to-one wire-matching scorer was run, and the human wire annotations are attached to the unmasked images, so wire-level F1 is not reported. The occlusion-off arm is not a proposed configuration and is not a replacement estimate for the pipeline’s benchmark performance.

For the completion safeguards, the shared-component (self-loop) guard fires on 18 of 31 images and rejects 38 candidate merges. Because the primary metric projects each net to distinct component pairs, disabling the guard leaves the scored pair sets unchanged: the guard’s protection against same-device-pin shorts is invisible to this metric and therefore unquantified here. Requiring a wire witness instead of the relaxed distance fallback changes predictions on 6 of 31 images (TP/FP/FN 413/37/71; micro-F1 0.884, macro-F1 0.907). These arms establish activation and paired effects; they do not rank the safeguards by safety.

TABLE 5. Join strategy and baseline comparison on human-verified net-level ground truth (31 real images, annotated component boxes; detected wires except for the stated perfect-wire reference). Primary metric is micro-averaged (pair-pooled) component-pair F1; macro (mean per-image) F1 is shown for reference.

Method	F1	Prec	Rec	F1 _{macro}
<i>Deterministic joins (ours and ablations)</i>				
Scale-rel. graph + completion (ours)	0.890	0.919	0.864	0.901
Rescue graph + completion (prior)	0.829	0.789	0.874	0.853
Scale-rel. graph (base)	0.816	0.929	0.727	0.838
Rescue graph (base)	0.787	0.811	0.764	0.813
Radius union-find (legacy)	0.667	0.698	0.638	0.674
<i>Classical deterministic baselines (identical detected wires)</i>				
Hough + proximity (best)	0.805	0.750	0.868	0.847
Connected-components (best)	0.624	0.965	0.461	0.611
<i>Conditional VLM diagnostic / annotated-wire reference</i>				
VLM (Claude Opus 4.8)	0.923	0.970	0.880	0.949
Scale-rel. + completion, perfect wires	0.890	0.913	0.868	0.916

GOLD = added / revised text

Synthetic and real-image results

31-image join replay matches saved per-image counts.

OPEN: results use annotated component boxes, not autonomous image-to-netlist correctness.

PROBLEMS / NO PROBLEMS

NO RERUN MISMATCH FOUND

The 31-image annotated-box join output matched saved per-image counts; autonomous accuracy was not shown.

SUGGESTED ACTION

Retain the annotated-box caveat and avoid end-to-end claims.

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COMMENTS / DECISION

to 0.916 (Table 3), so the wire detector accounts for essentially zero pair-weighted F1. The residual error is intrinsic join ambiguity on hand-drawn geometry, not wire detection, consistent with the 0.976 wire-detection F1 reported above (Table 1). The join degrades only gently with circuit size, with per-image F1 falling from 0.94 on circuits with ≤ 6 components to 0.86 above that and only weak correlation $r = -0.19$, and remains high-precision throughout, in contrast to the VLM, whose precision collapses on the densest circuits (Section IV-F).

E. PER-CIRCUIT ANALYSIS

Table 4 shows the scale-relative completion join’s performance across eight representative circuits from the 15-circuit synthetic corpus at L4 (extreme error injection, 16 seeds).

TABLE 4. Scale-Relative Completion Join Performance per Circuit at L4 (16 seeds)

Circuit	Comps	F1	Prec	Sim. acc.
Parallel R–R	3	1.00	1.00	25%
R–R divider	3	0.99	1.00	69%
RL series	3	0.99	1.00	81%
Diode–R loop	3	0.99	1.00	69%
Dual R loops	4	0.93	0.90	12%
4-component loop	4	0.94	0.91	56%
Wheatstone bridge	6	0.82	0.85	6%
6-component ring	6	0.95	0.92	50%

Precision remains high (≥ 0.85) across all circuits even at L4: errors cause fragmentation (dropped connections) rather than shorts (false merges). The Wheatstone bridge is the hardest topology (F1 = 0.82): its balanced arms make the inferred node structure sensitive to a single mis-placed connection, and simulation accuracy falls to 6%.

F. VLM CONNECTIVITY EXPERIMENT

To quantify the separation argued in Section II, that connectivity is better solved by an explicit geometric model than left to a VLM, we ran a first-party experiment with a strong general VLM (Claude Opus 4.8). To make the comparison fair, the VLM is given exactly what the join receives: the scanned image with the same ground-truth component boxes labelled by index, and is asked to return, as JSON, the sets of component terminals that share an electrical node; we score the resulting component-pairs with the same metric. Each image is read by an independent, context-free model call (one per image), so no answer leaks between circuits, and the human, not the model, remains the verifier of record. On a clean synthetic control (15 authored circuits) the VLM scores F1 = 0.99, confirming it understands the task and symbol set. On all 31 real scans, evaluated against the human-verified net-level ground truth, it

reaches micro-F1 = 0.923 (P = 0.97, R = 0.88; macro-F1 = 0.949).

The VLM thus edges our deterministic join on aggregate (0.923 vs 0.890 micro-F1), but the gap is not statistically significant: a paired bootstrap over the 31 images puts the difference at +0.033 with a 95% CI of $[-0.009, +0.078]$, which includes zero. Two further findings qualify the comparison. First, the VLM’s accuracy degrades with circuit complexity: it is exact (F1 = 1.00) on 21 of the 31 circuits, including all the small 3–5-component cases, but on the largest boards it systematically omits connections, with recall falling to 0.63–0.68 on the three densest boards, exactly where a netlist error is most costly; its precision otherwise stays high (0.97 micro), so its failure mode is dropped nets rather than shorts. The macro-F1 (0.949) is correspondingly inflated by the many easy small circuits, which is why we report the pair-weighted micro-F1 as primary. Second, the cost and form of the output differ from the geometric pipeline: each image consumes on the order of 10^5 tokens, the result is free-form text rather than a structured netlist, and nothing in the model guarantees electrical validity (e.g. that a two-terminal component is not shorted). Our geometric join is statistically indistinguishable from the VLM on these circuits while being deterministic, two to three orders of magnitude cheaper, structurally guaranteed against shorts by the self-loop guard, and simulatable by construction. We therefore treat the VLM as an upper reference for what a frontier model can do here, not as a connectivity module to ship: an explicit model reaches the same accuracy and also gives the determinism, low cost, and structural validity a deployed digitizer needs.

V. DISCUSSION

The endpoint-graph model resolves ambiguity that radius-based methods cannot: by representing both wire endpoints and component pins as nodes with typed edges, it distinguishes between a wire that passes near a component (endpoint–wire-body edge) and one that connects to it (endpoint–pin edge with directional preference). The component-first assignment further disambiguates by using the geometric constraint that wires must reach a component’s pin, not just its bounding box.

Degree-budget completion addresses the fundamental limitation of detector noise: even a 97.6% accurate wire detector drops ~2% of wire endpoints, and those dropped endpoints are precisely the connections that matter most for simulation accuracy. The b-matching formulation bounds the damage from mis-detection (each floating pin gets at most one recovery edge), while the self-loop guard prevents the catastrophic short-circuit failures that plague unconstrained approaches.

The human-verified net-level evaluation, while novel for hand-drawn circuits, currently covers 31 images; the

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PROBLEMS / NO PROBLEMS

OPEN EVIDENCE GAP

Crossover detector localization lacks a committed per-instance output record.

SUGGESTED ACTION

Add inputs, matching rule and per-instance output, or narrow the specific numerical claim.

Attribution note: Pranavesh authored earlier benchmark commits. Git lists Bosco Chanam as author of the later manuscript revision commits; confirm individual edit credit with the authors.

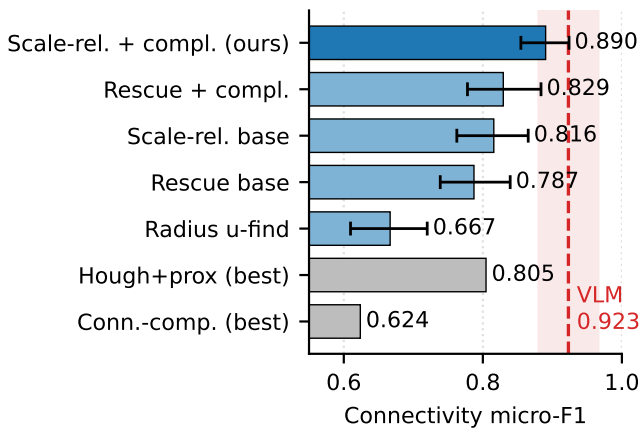


FIGURE 7. Micro-averaged connectivity F1 on the 31-image human-verified net-level benchmark (component-pair metric). The scale-relative completion join has the highest score among the evaluated deterministic configurations and classical baselines on these detected wires. The dashed line marks the VLM diagnostic (Claude Opus 4.8). Both methods receive annotated component boxes; the VLM-minus-join difference is +0.033 (95% bootstrap CI [-0.009, +0.078]). The interval does not establish equivalence, cost superiority or a global validity guarantee. Error bars are 95% bootstrap CIs.

This real-image ranking resolves an open question from prior work on this pipeline: against bootstrapped pilot labels the degree-budget completion had *not* consistently beaten its base join; against human-verified ground truth it clearly does (micro-F1 0.829 vs 0.787 for the rescue base alone), and the scale-relative variant is better still. The same ordering holds on the authored synthetic suite (Table 2), where the scale-relative completion join also ranks first, providing corroborating evidence on independently authored data without ruling out residual bias in the real benchmark labels.

Several image-to-netlist systems erase component regions and infer nets from the remaining wire pixels using connected components or contours, including SINA [3] and Peker et al. [5]. We evaluate a connected-component baseline on our detected wire segments to control the wire input. Its best tested configuration reaches component-pair micro-F1 0.624, versus 0.890 for the endpoint-graph join (Table 5). Hough line fitting [20] followed by proximity linking reaches 0.805 at its best tested linking tolerance on the same detected wires. These are the best configurations in the reported sweeps, not upper bounds on all implementations of the classical methods. Their ordering is evidence for these configurations and this benchmark, not a direct performance comparison with the cited complete systems.

We additionally attempted to run five public image-to-netlist systems on the hand-drawn benchmark and encountered the following implementation, model-vocabulary and input-domain limitations. The classical printed-schematic interpreter of Kelly and Cole [2] installs and runs, but on hand-drawn scans it over-segments wires into spurious components and recovers almost no connectivity, a domain-transfer failure. Netlistify [27] reproduces on its own printed AMS test data, which we ran end-to-end with the released model, but

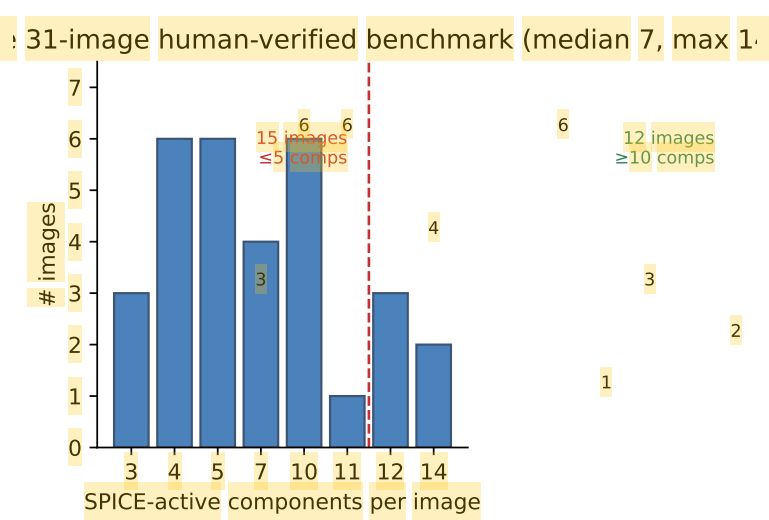


FIGURE 8. Complexity of the 31-image human-verified benchmark, counting components in the evaluated electrical subset: 15 images have ≤ 5 and 12 have ≥ 10 (median 7, range 3–14). Component-count breadth does not validate dense buses or multilayer crossings; performance on these cases remains untested.

its active model lacks the diode, inductor, and IC classes our circuits contain and its connectivity model is trained on printed thin-line appearance, limiting its applicability to the device population in this benchmark. Masala-CHAI [4] delegates connectivity to a proprietary LLM behind a paid API, equivalent in spirit to our VLM experiment in Section IV-H, rather than a deterministic algorithm. SINA’s [3] anonymized code archive was unextractable and CircuitNet [28] ships only Colab notebooks without released weights. These observations concern the versions and configurations examined; they do not establish that no prior system can recover hand-drawn connectivity. The CCL and Hough experiments above provide controlled comparisons on the same detected wires.

With annotated component boxes held fixed, replacing detected wires by the dataset’s annotated wires changes component-pair micro-F1 from 0.8903 to 0.8898, while macro-F1 increases from 0.901 to 0.916 (Table 5). The micro scores round to the same displayed value, but are not identical. This comparison concerns wire input conditional on component annotations; it does not establish an autonomous detection bottleneck or exclude compensating changes in recovered connections.

F. DESCRIPTIVE PER-DRAFTER RESULTS

Table 6 reports the scale-relative completion join grouped by CGHD drafter. The published CGHD-1152 dataset file index (version 14) places each of the 31 benchmark image filenames under exactly one drafter_N/images/ path, matching the drafter assignments used here. Thus all 31 assignments are direct matches to the corpus’s drafter labels rather than inferences from circuit numbering; the version 13 image-path index agrees. Four groups contain at least four images, with micro-F1 ranging from 0.854 to 0.987. These groups describe

GOLD = added / revised text

Ablations and uncertainty

Response adds bounded mechanism/edge ablations.

OPEN: crossover detector claims need raw per-instance evidence or narrower wording.

PROBLEMS / NO PROBLEMS

OPEN EVIDENCE GAP

Crossover detector localization lacks a committed per-instance output record.

SUGGESTED ACTION

Add inputs, matching rule and per-instance output, or narrow the specific numerical claim.

YOUR REVIEW STATUS

CHANGE OWNER / CONTRIBUTOR

COMMENTS / DECISION

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95% bootstrap confidence intervals on the per-strategy F1 are correspondingly wide (± 0.03 – 0.05) and overlap between adjacent strategies, so while the top ranking is consistent and corroborated by the independently authored synthetic suite, broader annotation is needed to separate the closely ranked configurations. The net-level labels were bootstrapped by a conservative endpoint-graph join over perfect wires and then human-corrected; we mitigate the residual concern that this favors graph-based methods by validating on synthetic ground truth that is authored from scratch, where the same ranking holds. The evaluation draws from a single hand-drawn corpus (CGHD-1152); generalization across other drawing styles, capture conditions, and printed schematics remains to be validated, as does extension to circuits containing devices outside the simulatable set (switches, transformers, etc.), which we exclude from the net-level benchmark. The VLM connectivity comparison uses a single frontier model (Claude Opus 4.8) on the same 31 images; evaluating additional models and prompting strategies is future work. Component detection at 88.5% mean average precision (mAP@0.5) also leaves room for improvement, particularly on crossovers (70.7% recall). The component-detection model was trained on CGHD-1152 after excluding drafter_0, whose drawing style diverged markedly from the other drafters; this exclusion reduces the training set but avoids domain-shift artifacts from an outlier style. Finally, the synthetic error model is a first-pass approximation of detector noise rather than a calibrated fit; with the new real-image net-level metric available, calibrating it to the measured detector failure distribution is a natural next step.

VI. CONCLUSION

We present a complete deterministic pipeline for hand-drawn circuit digitization, built on an occlusion-first wire extractor that yields an endpoint representation and a two-stage endpoint-graph join, together with, to our knowledge, the first human-verified net-level connectivity benchmark for hand-drawn circuits. The endpoint-graph join with degree-budget completion is most robust under synthetic error injection (F1 = 0.95 at the highest severity vs. 0.36 for the radius baseline), and, paired with a high-precision scale-relative base, attains connectivity micro-F1 = 0.890 on real images against human-verified ground truth, beating the prior completion default (0.83) and every deterministic baseline, including the connected-component net-tracing recipe used by recent systems (0.62) and a Hough-plus-proximity line tracer (0.81), all on identical detected wires. Running the join on perfect wires does not raise micro-F1 at all, showing the join, not wire detection, is the bottleneck. A first-party experiment with a frontier VLM reaches 0.923 on the same images, a difference from our join that is not statistically significant (paired 95% CI $[-0.009, +0.078]$), but at two to three orders of

magnitude higher cost and without simulatable, structurally valid output, supporting the explicit geometric design. Future work includes scaling the human-verified benchmark, calibrating the synthetic error model to the measured detector distribution, and extending coverage to devices outside the simulatable set.

ACKNOWLEDGMENT

The authors used Claude Opus 4.8 (Anthropic) to assist with grammar and language editing of the manuscript and as a programming assistant during development of the pipeline and evaluation code. This assistance is distinct from the VLM connectivity experiment in Section IV-F, where the same model is an evaluation subject. All methods, experiments, and conclusions are the authors' own, and the authors reviewed and take full responsibility for the entire content of this article.

DATA AND CODE AVAILABILITY

The pipeline, evaluation scripts, synthetic-circuit generator, the 31-image human-verified net-level ground truth (ground_truth/real_nets_verified.json), the net-level verification tool used to produce it, the join-strategy and connected-component baseline benchmarks, and the Claude-VLM connectivity harness are available at <https://github.com/boscochanam/circuit-digitization>. The component-detection model is available at <https://huggingface.co/boscochanam/circuit-component-detector>, trained on the CGHD-1152 dataset [1].

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PROBLEMS / NO PROBLEMS

OPEN AUTHOR CHECK

34-to-31 exclusions await author sign-off; detector training overlap with scored images is unknown.

SUGGESTED ACTION

Confirm exclusions from source records; avoid any held-out claim without a split manifest.

Attribution note: Pranavesh authored earlier benchmark commits. Git lists Bosco Chanam as author of the later manuscript revision commits; confirm individual edit credit with the authors.

within-corpus variation, not evidence of unseen-drafter transfer. No held-out-drafter experiment was performed; the small groups do not support strong statistical separation.

TABLE 6. Descriptive join micro-F1 (scale-relative completion) by CGHD drafter. Groups with at least four images; overall includes all benchmark images. Drafter identities are direct filename matches to the published CGHD-1152 file index (31/31).

Drafter	<i>n</i> images	micro-F1	macro-F1	TP/FP/FN
drafter_1	7	0.933	0.907	56/6/2
drafter_2	5	0.854	0.874	85/14/15
drafter_10	5	0.987	0.995	37/1/0
drafter_12	4	0.913	0.938	21/0/4
Overall	31	0.890	0.901	418/37/66

G. PER-CIRCUIT ANALYSIS

Table 7 shows the scale-relative completion join’s performance across eight representative circuits from the 15-circuit synthetic corpus at L4 (extreme error injection, 16 seeds).

TABLE 7. Scale-Relative Completion Join Performance per Circuit at L4 (16 seeds)

Circuit	Comps	F1	Prec	Sim. acc.
Parallel R–R	3	1.00	1.00	25%
R–R divider	3	0.99	1.00	69%
RL series	3	0.99	1.00	81%
Diode–R loop	3	0.99	1.00	69%
Dual R loops	4	0.93	0.90	12%
4-component loop	4	0.94	0.91	56%
Wheatstone bridge	6	0.82	0.85	6%
6-component ring	6	0.95	0.92	50%

Precision is at least 0.85 across the reported synthetic circuits at L4. This does not exclude false merges or establish that all errors are fragmentation. The Wheatstone bridge has the lowest reported F1 (0.82) and simulation accuracy of 6%, illustrating that component-pair connectivity and the separately evaluated synthetic simulation criterion measure different properties.

H. VLM CONNECTIVITY EXPERIMENT

We evaluate Claude Opus 4.8 as an oracle-component-box connectivity diagnostic. Both methods receive the same annotated component boxes; the geometric method uses detected wires. The VLM is given the scanned image with component boxes labelled by index and asked to return JSON sets of component terminals sharing an electrical node. Each image uses a separate model call without shared conversational context. We score the resulting unordered component pairs against the same human-verified benchmark. This is a conditional connectivity comparison, not an autonomous end-to-end comparison.

On the 31 real images, the VLM reaches micro-F1 0.923 (precision 0.970, recall 0.880; macro-F1 0.949), versus micro-F1 0.890 for the geometric join, and is exact under the component-pair metric on 21/31 images. The paired

VLM-minus-join difference is +0.033, with 95% bootstrap CI [−0.009, +0.078]. Inclusion of zero does not establish equivalence. These results do not establish comparative cost superiority or a global guarantee against incorrect merges for either method. Autonomous performance requires separately auditable evaluation; the conditional scores do not certify pin-level or simulation correctness.

V. DISCUSSION

The endpoint graph represents wire endpoints and component pins with separate rules for endpoint proximity, component assignment and mid-span contacts. Component-first assignment selects a nearby component by OBB distance and then a pin by Euclidean distance. These geometric choices provide inspectable connectivity candidates, but do not establish that a nearby wire is electrically connected or that a selected pin has the correct device-specific meaning.

Degree-budget completion proposes connections for pins left in single-component nets. Each floating-pin assignment selects at most one recovery target, and the shared-component guard can reject additional merges. Neither mechanism repairs shorts already present in the base graph. Wire-detection F1 does not measure an endpoint-drop rate or establish the causal contribution of missing endpoints to simulation failures.

The human-verified net-level evaluation, while novel for hand-drawn circuits, currently covers 31 images; the 95% bootstrap confidence intervals on the per-strategy F1 are correspondingly wide (±0.03–0.05) and overlap between adjacent strategies, so while the top ranking is consistent and corroborated by the independently authored synthetic suite, broader annotation is needed to separate the closely ranked configurations. The net-level labels were bootstrapped by a conservative endpoint-graph join over perfect wires and then human-corrected; we mitigate the residual concern that this favors graph-based methods by validating on synthetic ground truth that is authored from scratch, where the same ranking holds. Annotation was performed by a single human annotator using a purpose-built verification tool, and no per-image timing log was retained, so a per-image annotation-cost figure cannot be reported. A proposed expansion protocol would stratify sampling by component count, drafter, crossing/bus structure and capture conditions, use independent annotation followed by adjudication, and record per-image effort and provenance logs. This is a proposed expansion protocol, not completed annotation. The evaluation draws from a single hand-drawn corpus (CGHD-1152); generalization across other drawing styles, capture conditions, and printed schematics remains to be validated.

Table 8 distinguishes the evaluated connectivity scope from generic pin guesses and illustrative export. Listing a device group does not establish separately detected model classes: the detector vocabulary merges multiple original classes. Pin construction depends on the component type supplied to the netlist stage; a template for a named complex device does not recover that identity from a merged detector label. Generic

GOLD = added / revised text

Drafter and generalization limits

Within-corpus subgroup mapping is documented.

OPEN: unknown detector train/test overlap; 34 annotated to 31 scored needs author sign-off.

PROBLEMS / NO PROBLEMS

OPEN AUTHOR CHECK

34-to-31 exclusions await author sign-off; detector training overlap with scored images is unknown.

SUGGESTED ACTION

Confirm exclusions from source records; avoid any held-out claim without a split manifest.

YOUR REVIEW STATUS

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COMMENTS / DECISION

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BOSCO CHANAM received the B.Tech. degree in Computer Science and Engineering from the Symbiosis Institute of Technology (SIT), Pune, India. He is currently an AI/ML Engineer at the AI Incubation Centre at Bharat Forge (Kalyani Group), Pune, India. His professional and research interests include computer vision, edge AI optimization, and robotics. Mr. Chanam has co-authored research in natural language processing benchmarking, published in the International Journal of Fuzzy Logic and Intelligent Systems.



CHRIS DCOSTA received the B.Tech. degree in Computer Science and Engineering with Honors in Artificial Intelligence and Machine Learning from the Symbiosis Institute of Technology (SIT), Pune, India. His research interests include computer vision, machine learning, natural language processing, and deep learning.



PRANAVESH KUMAR TALUPURI is with the Department of Computer Science, University of Southern California, Los Angeles, CA, USA. His research interests include machine learning, deep learning, and computer vision.



SHWETAMBARI CHIVHANE holds a B.E. in Computer Science and Engineering from Sant Gadge Baba Amravati University, Amravati, India, and an M.Tech. in Computer Science and Engineering from Dr. Babasaheb Technological University, Lonere, India. She earned her PhD in Computer Engineering specializing from Bharath Institute of Higher Education and Research, Chennai, India. Presently, she serves as an Assistant Professor in the Department of Computer Science and Engineering at Symbiosis Institute of Technology, Symbiosis International (Deemed University), Pune, India, gathering over two decades of teaching and research expertise. She is life member of ISTE.

Her contributions extend beyond academia; she has contributed as a reviewer for various international conferences. Dr. Shwetambari’s research expertise is evident with her receiving many publications. She has authored over 35 research articles in various international and national conferences and journals. Her professional engagements include roles such as NAAC, NBA, among others.

Recognized as a PhD guide in Symbiosis International (Deemed University), Presently she is guiding 4 research scholar and she has guided more than 100 undergraduate students for their research endeavors. Her current research interests revolve around Image Processing, Computer Vision, Artificial Intelligence and Machine Learning.

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LIMITATIONS DISCLOSED

Extreme-size, mixed-size and simulation-equivalence claims are not experimentally established.

SUGGESTED ACTION

Keep them explicit limitations; do not convert them into demonstrated robustness.

Attribution note: Pranavesh authored earlier benchmark commits. Git lists Bosco Chanam as author of the later manuscript revision commits; confirm individual edit credit with the authors.

pin geometry is not evidence of validated device-specific pin relationships.

Component detection has released-checkpoint mAP@0.5 of 89.0% (the checkpoint embeds 0.88977 at its best-fitness epoch; the previously reported 88.5% belongs to the final training epoch), with crossover recall of 70.7%. Geometric edge types 2, 4 and 5 do not consult crossover class labels when joining nearby wire fragments or mid-span contacts, creating a possible false-merge pathway. An audit using annotated component boxes found 13 crossover instances across eight benchmark images. In C242_D1_P1, suppressing two crossing-adjacent endpoint–endpoint edges changed component-pair TP/FP/FN from 27/4/0 to 19/0/8. Removing these edges also removes true connections; this is not a demonstrated mitigation gain. To test the detector half of this pathway directly, we ran the trained detector over the 31 benchmark images and compared its crossover outputs with the 13 annotated instances. Every annotated crossover was localized and assigned to the crossover class at a confidence of at least 0.25 (confidence 0.28–0.97, IoU 0.43–0.91); at a 0.50 threshold, 12 of 13 are retained and one falls below threshold. This does not contradict the reported 70.7% class recall, which is measured on the detector’s own validation data at its deployment threshold, and training-split membership for these benchmark images is unknown, so these are not held-out estimates. As a paired counterfactual, deleting the annotated crossover boxes before joining changes the recovered predictions on 6 of 31 images and changes component-pair micro-F1 by +0.010, with identical net partitions on 23 of 31 images. These two conditional checks therefore give no evidence that crossover misclassification is the driver of the false pairs observed here. Both are annotated-box, within-benchmark diagnostics: they are not a prevalence estimate, not an end-to-end error rate, and not a validation of crossover-aware suppression, which remains unvalidated.

The VLM comparison uses a single model (Claude Opus 4.8) on 31 images; evaluating additional models and prompting strategies is future work. The component detector was trained on CGHD-1152 after excluding drafter_0, whose drawing style differed from the retained drafters; performance on that excluded style is not established here. The synthetic error model is an authored approximation rather than a calibrated fit to detector errors. Calibration to measured errors is future work.

VI. CONCLUSION

We present structural circuit-netlist recovery using an occlusion-first wire extractor and endpoint-graph joining with degree-budget completion. On the 31-image within-corpus benchmark with annotated component boxes, component-pair micro-F1 is 0.8903; perfect-wire input gives 0.8898. These scores round alike but do not establish an autonomous detection bottleneck. The component-pair metric does not certify pin identities, exact net partitions, absence of shorts, or simulation equivalence. Synthetic L4 connectivity F1 is 0.95 versus 0.36 for the radius baseline. The VLM compar-

ison is conditional on annotated component boxes and does not establish equivalence or cost superiority. Export remains illustrative: values and device models require external specification, and real-scan simulation equivalence is unvalidated. Broader styles, dense buses, and mixed-scale schematics require future evaluation.

ACKNOWLEDGMENT

The authors used Claude Opus 4.8 (Anthropic) to assist with grammar and language editing of the manuscript and as a programming assistant during development of the pipeline and evaluation code. This assistance is distinct from the VLM connectivity experiment in Section IV-H, where the same model is an evaluation subject. All methods, experiments, and conclusions are the authors’ own, and the authors reviewed and take full responsibility for the entire content of this article.

DATA AND CODE AVAILABILITY

The pipeline, evaluation scripts, synthetic-circuit generator, the 31-image human-verified net-level ground truth (ground_truth/real_nets_verified.json), the net-level verification tool used to produce it, the join-strategy and connected-component baseline benchmarks, and the Claude-VLM connectivity harness are available at <https://github.com/boscochanam/circuit-digitization>, including the per-image to CGHD drafter mapping used for the per-drafter analysis (Table 6). The component-detection model is available at <https://huggingface.co/boscochanam/circuit-component-detector>, trained on the CGHD-1152 dataset [1].

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GOLD = added / revised text

Discussion and limitations

Best-checkpoint metric distinguished from final epoch.

OPEN: no extreme-size or controlled mixed-size experiment; no simulation-equivalence proof.

PROBLEMS / NO PROBLEMS

LIMITATIONS DISCLOSED

Extreme-size, mixed-size and simulation-equivalence claims are not experimentally established.

SUGGESTED ACTION

Keep them explicit limitations; do not convert them into demonstrated robustness.

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Chanam *et al.*: From Hand-Drawn Schematics to SPICE Netlists



ASHAY KUMAR SINGH received the B.Tech. degree in Computer Science and Engineering from the Symbiosis Institute of Technology (SIT), Pune, India. His research interests lie in machine learning, computer vision, Retrieval-Augmented Generation (RAG), optical character recognition (OCR), and multimodal AI.



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SUGGESTED ACTION

Validate references, author details and response-to-reviewers before submission.

Attribution note: Pranavesh authored earlier benchmark commits. Git lists Bosco Chanam as author of the later manuscript revision commits; confirm individual edit credit with the authors.

TABLE 8. Connectivity evaluation, generic pin construction, and illustrative export are distinct capabilities.

Device group	Evaluated scope	Pin construction	Export limitation
R/C/L/D/Q, voltage sources and ICs	Component-pair connectivity on the benchmark; no universal validation of pin templates	Geometric templates; endpoint clustering for eligible discrete/source types, generic layouts for ICs	Illustrative defaults/substitutions; values and device models require external specification. ICs are skipped because no export model is implemented.
Switches	Absent from the evaluated electrical subset; switch-specific connectivity unvalidated	Two pins from the shorter OBB-edge midpoints; long-axis AABB fallback	Fixed 0.001 Ω resistor if pins 0 and 1 map to distinct nodes; otherwise no switch line. No switch state or control model is recovered.
Transformers, thyristors, optocouplers and other complex devices	Device-specific connectivity unvalidated	Named transformer type receives a two-pin geometric guess; optocoupler template and unknown-type fallback also exist	These complex types have no implemented export model and are skipped; geometric guesses do not establish winding, control or isolation semantics.
Metric boundary (all evaluated groups)	Component pairs discard terminal identity	F1 does not certify pin assignment or the exact net partition	F1 does not certify absence of shorts or simulation equivalence.

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Fuzzy Logic and Intelligent Systems.

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CHRIS DCOSTA received the B.Tech. degree in Computer Science and Engineering with Honors in Artificial Intelligence and Machine Learning from the Symbiosis Institute of Technology (SIT), Pune, India. His research interests include computer vision, machine learning, natural language processing, and deep learning.



PRANAVESH KUMAR TALUPURI is with the Department of Computer Science, University of Southern California, Los Angeles, CA, USA. His research interests include machine learning, deep learning, and computer vision.

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Capabilities and references

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SHWETAMBARI CHIWHANE holds a B.E. in Computer Science and Engineering from Sant Gadge Baba Amravati University, Amravati, India, and an M.Tech. in Computer Science and Engineering from Dr. Babasaheb Technological University, Lonere, India. She earned her PhD in Computer Engineering specializing from Bharath Institute of Higher Education and Research, Chennai, India. Presently, she serves as an Assistant Professor in the Department of Computer Science and

Engineering at Symbiosis Institute of Technology, Symbiosis International (Deemed University), Pune, India, gathering over two decades of teaching and research expertise. She is life member of ISTE.

Her contributions extend beyond academia; she has contributed as a reviewer for various international conferences. Dr. Shwetambari's research expertise is evident with her receiving many publications. She has authored over 35 research articles in various international and national conferences and journals. Her professional engagements include roles such as NAAC, NBA, among others.

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